# Green Customer Segmentation And Retention Analysis




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# Abstract

Companies often assume that customers who buy sustainably are also the loyal ones, but that link is rarely put to an actual test. This study does exactly that, using a real-world dataset of 2,212 retail customers, by combining customer segmentation with a model investigating purchase inactivity. RFM features and a Green_Ratio metric (an observed spending-based proxy for green preference, not a confirmed measure of environmental motivation) were engineered from the data; an early inactivity label was found to leak information and was redefined using Recency alone, so the resulting model investigates associations with a 90-day inactivity status rather than predicting future churn outright. K Means clustering, confirmed through the elbow method, silhouette analysis, and stability checks, produced five behavioral segments, while a tuned XGBoost classifier, benchmarked against logistic regression, was trained on income, age, green ratio, household size, and tenure, with SHAP used to examine its reliance on each feature. Under the 90-day threshold, 8.8% of customers were classified as inactive, and no significant relationship emerged between green spending and inactivity across a chi-square test, a logistic regression, and a nonlinearity check (all $p > 0.68$); the model itself showed only marginal discriminative power (test AUC = 0.527). SHAP's apparent reliance on Green_Ratio did not hold up under a multivariate logistic regression controlling for income, age, household size, and tenure, since none of these variables reached significance either, arguing against confounding and instead suggesting the model was picking up weak, non-generalizable noise. The clustering still surfaced useful personas, including a high-recency, high-inactivity segment and a younger, high green-share segment, and together these results suggest green marketing and retention marketing are best pursued as separate levers, with recency, not sustainability signals, driving win-back efforts.

**Keywords:** Customer Segmentation, Churn Prediction, Sustainable Consumption, K-Means Clustering, XGBoost, Explainable AI, SHAP, Data Leakage.

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# Chapter 1 Introduction

## 1.1 Background

Most retailers today have access to more customer data than they are able to fully exploit, and two questions tend to dominate how that data is used: who are the organisation's customers, and which of them are at risk of being lost? The first question falls within the domain of customer segmentation, which groups customers according to their behaviour so that marketing efforts can be tailored to each group rather than applied uniformly. The second falls within the domain of churn prediction, which attempts to identify customers who are disengaging before the relationship is lost. Neither concept is new, but both remain central to marketing analytics for a straightforward reason: retain an existing customer is consistently less costly than winning back a lost one (Ekene et al. 2024).

A further, more recent pressure on this landscape is sustainability. Consumers increasingly report that environmental impact factors into their purchasing decisions; globally, approximately four in ten consumers surveyed since 2021 report purchasing at least one sustainable product every month (Rogers & Cascone 2025). This shift is increasingly difficult for retailers to disregard. Within the European Union specifically, regulation is placing direct pressure on companies: the Corporate Sustainability Reporting Directive requires large companies to disclose detailed environmental impact data (European Union 2022), and the Green Claims Directive and the Directive on Empowering Consumers for the Green Transition require firms to substantiate any "green" marketing claim with verifiable evidence or risk penalties for greenwashing (de Lange et al. 2024). Retailers additionally face increasing expectations from investors and financing partners, who increasingly incorporate sustainability performance into lending and investment decisions (de Lange et al. 2024). Sustainability adoption is consequently being driven by regulators, capital markets, and consumers simultaneously, rather than remaining a matter of discretionary corporate pace.

A commonly stated assumption follows from this context: that customers who purchase more sustainably are also more loyal, allowing green marketing and retention marketing to be treated as a single combined effort. This assumption has rarely been tested directly. This thesis examines it empirically, using a real-world retail dataset of just over two thousand customers, through a pipeline combining customer segmentation, churn prediction, and explainable artificial intelligence, to assess whether sustainability-oriented purchasing behaviour and customer retention are genuinely associated. This study operationalises *churn* as a 90-day purchase-inactivity proxy rather than observed future attrition, and *green behaviour* as a fruit-and-fish spending proxy rather than confirmed sustainable purchasing; both constructs, and the reasoning behind them, are discussed in Chapter 4.

## 1.2 Problem Statement

Two gaps characterise how this question is addressed in practice. First, the claim that sustainability-minded customers are more loyal is frequently accepted without direct testing, or is supported with survey data describing what consumers report care about; stated attitudes are a well-documented poor predictor of actual purchasing behaviour (ElHaffar et al. 2020). Second, and independently, churn-prediction research often omits a methodologically important step: verifying that the churn label itself was not constructed using variables later reintroduced as model predictors (Vo et al. 2026). Where this occurs, a model can achieve seemingly strong performance without capturing genuine behavioural signal, since it is effectively reconstructing its own label. This thesis addresses both gaps: it constructs a churn definition that has been audited for this form of leakage, and only once that foundation is established does it examine the substantive question of whether green spending is associated with customer retention.

## 1.3 Research Objectives

The aims of this study are to:

1. Engineer RFM (Recency, Frequency, Monetary) features, along with a Green Ratio metric capturing how much of each customer's spending goes toward lower-footprint products

2. Construct a 90-day purchase-inactivity proxy that is free of data leakage — including fixing an earlier version that wasn't, and explicitly distinguishing this proxy from observed future churn.

3. Group customers into a manageable number of interpretable segments using K-Means clustering

4. Train and tune a model (XGBoost) to investigate associations with the 90-day inactivity proxy, with a logistic regression baseline as a point of comparison.

5. Use SHAP to examine the trained model and assess whether green-spending behaviour is meaningfully associated with its predictions of purchase inactivity.

6. Turn the segmentation and inactivity-proxy findings into practical, segment-specific marketing recommendations.

## 1.4 Research Questions

- RQ1: Do meaningful, distinct customer segments emerge from RFM and green spending behavior?

- RQ2: Once the inactivity proxy is leakage free, is green spending behaviour considerably associated with 90 day purchase inactivity?

- RQ3: How well can 90 day purchase inactivity be discriminated from the remaining features, and does a tuned model suggestively outperform a simple baseline?

- RQ4: What does SHAP reveal about which features the model relies on most — and does that support or undercut the raw descriptive relationship between green-spending behaviour and 90-day purchase inactivity?

- RQ5: What does all of this suggest for how a business should actually treat each customer segment?

## 1.5 Contributions

- A worked example of grabbing and fixing data leakage in a purchase-inactivity label — not just mention leakage as a risk, but showing what it looked like here and how it was resolved.

- A Green Ratio metric built entirely from transaction data, the usual reliance on self-reported sustainability attitudes

- Five named, business-usable customer segments, grounded in actual cluster behavior rather than assumed in advance.

- An evaluation approach that doesn't let an optimistic cross-validation score stand in for real performance — the held-out test result is treated as the number that counts.
- A SHAP-based exploratory analysis of why the model assigns some importance to green-spending behaviour despite the absence of a strong raw correlation with the 90-day inactivity proxy, interpreted cautiously given the model's limited discriminative performance.

## 1.6 Thesis Organization

The rest of the thesis is organized as follows. In Chapter 2, the background of the study is presented, laying the groundwork by discussing the theoretical foundations and context in which the research is situated such as segmentation, churn, the machine-learning tools used here, and what's known about green consumer behavior.Chapter 3 provides a comprehensive literature review, critically examining previous work and highlight's gaps that this research aims to address. In Chapter 4, walks through the methodology in full: cleaning, feature engineering, the churn-label fix, clustering, model training, and explainability. Chapter 5 presents the results and discussions, where the findings are analyzed in relation to the research objectives. Finally, Chapter 6 closes with a summary, the study's limitations, and where this work could go next.

# Chapter 2 Background

This chapter introduces the concepts underpinning this thesis. None of these concepts originate with this study; they are drawn from established marketing analytics and machine learning research. Understanding them here provides the necessary foundation for the methodology described in Chapter 4 and clarifies where this thesis extends existing practice rather than restating it.

## 2.1 Customer Segmentation and RFM Analysis

Customer segmentation addresses a practical constraint in marketing: customers differ in their needs and behaviour, so treating a customer base uniformly is inefficient. Segmentation divides a customer base into groups that are internally similar and externally distinct, enabling more targeted messaging, offers, and retention effort.

One of the most established segmentation frameworks is RFM analysis (Recency, Frequency, Monetary), which reduces a customer's transaction history to three quantities: how recently a purchase was made, how frequently purchases occur, and how much is spent (Suh 2025). The framework originated in direct-mail marketing and has remained in use because it is computationally inexpensive and applicable to almost any transactional dataset. Subsequent work demonstrated that recency and frequency jointly carry most of the predictive signal regarding a customer's future value, formally linking RFM to the broader concept of customer lifetime value (Fader et al. 2005). This thesis uses RFM as its starting point, computing Recency, Frequency, and Monetary value directly from the dataset as the foundation for the remaining feature set.

## 2.2 Customer Churn and Why Retention Matters

Churn refers to customers ceasing to engage with a business. Substantial analytical effort is directed toward predicting churn because retaining an existing customer is

consistently less costly than replacing one, giving early identification of disengagement direct financial value.

Churn can be operationalised differently depending on the business context. A subscription-based business may define churn as account cancellation; a retailer without a formal membership structure must instead define churn behaviourally, typically through a threshold on time elapsed since the customer's last purchase. This threshold, and the variables permitted to inform it, materially affect the resulting analysis: an inappropriately constructed definition can render the prediction problem trivial or, more seriously, can introduce leakage into the label itself. This issue is addressed directly in Chapter 4, where the churn definition adopted in this thesis required correction for precisely this reason.

## 2.3 K-Means Clustering

K-Means remains one of the most widely used unsupervised learning algorithms and a common default for segmentation tasks (Ling & Weiling 2025). Given a specified number of groups, $k$, the algorithm iteratively assigns each observation to its nearest centroid, recomputes centroids based on current assignments, and repeats until assignments stabilise.

Because K-Means relies on Euclidean distance, it is sensitive to the relative scale of input variables: a variable measured in the tens of thousands, such as income, can dominate the distance calculation relative to a variable bounded between zero and one, such as a spending ratio, even where the latter carries comparable analytical importance. Standard practice therefore standardises all variables prior to clustering.

Selecting $k$ requires further consideration, since no single objectively correct value exists; this is typically approached using two complementary diagnostics. The elbow method examines the rate at which within-cluster variance (inertia) decreases as $k$ increases, identifying the point at which further increases yield diminishing returns. The silhouette score quantifies how well-separated and internally cohesive the resulting clusters are for a given $k$ (Shahapure & Nicholas 2020). Neither diagnostic provides a definitive answer independently, and, as demonstrated in Chapter 4, the statistically optimal value of $k$ is not always the most useful one for a business audience.

## 2.4 Gradient-Boosted Trees: XGBoost

Whereas K-Means groups customers without reference to a target label, churn prediction is a supervised problem: past outcomes are known, and the objective is to learn a generalisable pattern. XGBoost has become one of the most widely adopted algorithms for structured, tabular prediction tasks of this kind, largely because it achieves strong performance without extensive manual tuning (Shwartz-Ziv & Armon 2021).

The algorithm constructs a sequence of decision trees, where each successive tree is trained to correct the residual errors of the ensemble built so far. A configuration parameter of particular relevance to this thesis is `scale_pos_weight`, which allows the model to weight the minority class more heavily than its raw frequency would otherwise suggest, an important consideration in churn problems, where customers who actually churn typically represent a small minority of the customer base, as is the case here.

## 2.5 Explainable AI and SHAP

A recognised trade-off exists in machine learning between predictive accuracy and interpretability. Simpler models, such as logistic regression, are inherently interpretable, since their coefficients can be read directly. More complex models, such as XGBoost, typically achieve stronger predictive performance but do not provide an equivalent, direct explanation of their predictions.

SHAP (SHapley Additive exPlanations) addresses this limitation. Drawing on the Shapley value from cooperative game theory, SHAP assigns each feature a contribution value for every individual prediction, such that these contributions sum exactly to the difference between that prediction and the model's average output (Lundberg & Lee 2017). This allows a trained model's predictions to be examined at the global level (aggregated across a dataset), at the segment level, or for a single individual prediction. This thesis draws on all three levels of SHAP output to assess a question central to the analysis: whether a feature the model relies on reflects a genuine, independent effect, or reflects an association with another, more directly influential variable.

## 2.6 Data Leakage

Data leakage occurs when information that would not legitimately be available at the point of prediction, or that is definitionally entangled with the outcome being predicted, is nonetheless incorporated into a model as a predictor (Gupta et al. 2025). Its effect is that a model can appear to perform strongly during development while failing to generalise once deployed, because the apparent pattern learned does not reflect genuine predictive signal.

Distinguishing leakage from two related but distinct concepts is important for the methodology in Chapter 4. *Leakage* occurs specifically when a predictor variable is used, directly or indirectly, to construct the target label itself; such a variable cannot be included as a predictor without the model partially reconstructing its own label. *Legitimate correlation*, by contrast, occurs when a variable is statistically associated with the outcome without having been used to define it; such variables remain valid predictors. *Temporal availability* is a separate consideration concerning whether a variable would have been observable at the point a prediction would legitimately be made, independent of whether it was involved in the label's construction. A common failure mode in churn modelling arises from defining the churn label using a rule that combines several variables, and subsequently including one of those same variables as a model predictor; the model can then partially reconstruct the labelling rule rather than learn a genuine behavioural pattern. This is precisely the issue identified and corrected in this thesis's churn-label construction, discussed in full in Chapter 4.

## 2.7 Sustainability and Green Consumer Behavior

Sustainability has become a mainstream consideration in retail rather than a peripheral one. Consumers increasingly report that sustainability factors into their purchasing decisions, regulatory frameworks increasingly expose unsubstantiated "green" marketing claims to legal risk, and investors are increasingly incorporating sustainability performance into their decision-making (de Lange et al. 2024, Rogers & Cascone 2025). Food production is particularly relevant to this discussion, since environmental footprint varies substantially across product categories; meat production, in particular, is associated with disproportionately higher greenhouse gas emissions, land use, and water use relative to plant-based or fish-based alternatives (Clark & Tilman 2017).

A well-documented gap nonetheless exists between consumers' stated sustainability attitudes and their actual purchasing behaviour, commonly referred to as the "green gap" or attitude-behaviour gap (ElHaffar et al. 2020, Adrita & Mohiuddin 2020): consumers frequently report stronger sustainability preferences than their purchasing behaviour reflects. This gap motivates this thesis's decision to avoid stated-attitude measures entirely, constructing its green-behaviour measure directly from observed spending data, and to treat any assumed relationship between green spending and customer loyalty as an empirical question requiring direct testing rather than an established premise.

# Chapter 3 Literature Review

Several of the individual components this thesis relies on — RFM-based segmentation, K-Means clustering, gradient-boosted churn models, SHAP-based explainability, and research on green consumer behaviour — have each been studied extensively in isolation. Considerably less common is work that combines these strands to test, rather than assume, whether sustainability-oriented spending is associated with customer loyalty. This chapter reviews the relevant literature in each of these areas, with the aim of identifying specifically where each strand falls short of addressing the question this thesis examines.

## 3.1 RFM-Based Customer Segmentation

RFM analysis has remained in use in customer analytics for decades, largely because it requires little input: three quantities, computable from almost any transaction log, that carry substantial predictive value (Huang 2025). Subsequent work formalised the basis for this, linking recency and frequency directly to customer lifetime value and demonstrating that these two dimensions alone carry most of the useful signal regarding a customer's future value (Fader et al. 2005).

This literature engages minimally with sustainability. RFM was constructed entirely around transactional value, how much, how often, how recently, without reference to what a customer purchases or whether this reflects environmental consciousness. This thesis addresses that gap directly, extending the RFM framework with a green-spending feature computed using the same transaction-based approach as RFM itself, rather than relying on survey data.

## 3.2 Clustering for Customer Segmentation

K-Means remains the default clustering algorithm for converting continuous RFM-style scores into discrete, usable segments (Kasem et al. 2023). Its principal limitation, that the number of clusters must be specified in advance, with no single

objectively correct value, is typically addressed by combining the elbow method with the silhouette score (Alamsyah et al. 2022). Applied segmentation studies rarely discuss explicitly that the statistically optimal number of clusters and the most business-useful number are often different values; most report whichever number the diagnostics favour without addressing this tension. This thesis treats that tension as worth discussing explicitly, selecting a cluster count that trades some statistical cohesion for practical marketing usefulness, and justifying that trade-off directly in Chapter 4.

## 3.3 Machine Learning for Churn Prediction

Churn prediction has been studied most extensively in subscription-based industries such as telecommunications, where a clear shift has occurred over time from simple statistical models toward ensemble methods such as gradient-boosted trees, which capture non-linear feature interactions more effectively with less manual engineering (Jain et al. 2020). Because churners typically represent a small minority of the customer base, this literature consistently identifies class imbalance as a central challenge, addressed through techniques such as class-weighting and threshold-independent metrics such as AUC rather than raw accuracy.

The gap here is methodological rather than conceptual: relatively few applied churn studies explicitly verify whether their churn label was constructed using variables that later reappear as model features. Leakage is acknowledged as a general risk within the data-mining literature (Kaufman et al. 2011), but it is rarely audited directly within specific applied churn case studies; most report performance figures without detailing how the label was constructed or whether that construction could be inflating results. This thesis treats this audit as a required step rather than an afterthought, demonstrating concretely what leakage looked like within this dataset and how correcting it altered the results.

## 3.4 Explainable AI in Predictive Modeling

As churn models have moved toward more powerful but less transparent algorithms, SHAP has become a standard tool for examining model behaviour, offering a mathematically grounded method for attributing each prediction to individual features (Lundberg & Lee 2017). It has been widely adopted in churn-modelling and marketing-analytics contexts to render model outputs actionable rather than opaque.

What remains largely absent from this literature is a rigorous use of SHAP to test a specific, substantive business hypothesis, rather than reporting a feature-importance ranking in isolation. Most applications stop at describing which features a model relies on, without examining why, or whether that reliance reflects a genuine relationship rather than a statistical artefact of a model with limited overall accuracy. This thesis extends this practice, using SHAP specifically to examine whether an apparent feature effect is genuine or reflects confounding, an application this literature rarely demonstrates.

## 3.5 Sustainability and Consumer Loyalty

The evidence that food categories carry substantially different environmental footprints is well established, with meat production identified as considerably more resource-intensive than plant-based or fish-based alternatives (Clark & Tilman 2017). Consumer-behaviour research has, separately, documented a persistent gap between stated sustainability attitudes and actual purchasing behaviour (ElHaffar et al. 2020). This "green gap" is typically examined through surveys of stated attitudes and intentions.

Direct, transaction-based evidence on whether green purchasing behaviour itself, rather than stated attitude, is associated with a downstream business outcome such as customer churn, remains notably limited. Existing work consists largely of attitudinal survey research without transactional grounding, or industry commentary asserting a green-loyalty relationship without empirical testing (de Lange et al. 2024, Rogers & Cascone 2025). This is the specific gap this thesis addresses.

## 3.6 Summaries of Related Studies

Drawing back from the individual studies reviewed above, several broader patterns emerge.

The segmentation literature has developed around RFM and K-Means as an established, low-cost combination (Fader et al. 2005). This combination is well understood and rarely the limiting factor in applied studies, but it was developed within a framework where "customer value" was defined by spending and frequency alone. Sustainability was not part of this original conception, and later applications have not substantially incorporated it.

The churn-prediction literature reflects a similar pattern of technical progress: a clear shift from simple linear models toward gradient-boosted ensembles such as XGBoost, which capture the non-linear complexity of customer behaviour more effectively (Jain et al. 2020). This progress has, however, been directed primarily toward improved accuracy rather than scrutiny of how that accuracy was obtained. Data leakage is well documented as a general risk (Kaufman et al. 2011), yet it is treated as a secondary consideration rather than a required checkpoint in most applied churn studies.

The explainability literature has provided a genuinely useful tool in SHAP (Lundberg & Lee 2017), but it is used predominantly to establish which features a model relies on, and is rarely extended to examine whether that reliance is substantively justified.

The sustainability literature, meanwhile, has extensively documented the divergence between consumers' environmental attitudes and their actual purchases (ElHaffar et al. 2020), while food-systems research has independently established that certain product categories carry a substantially greater environmental cost than others (Clark & Tilman 2017). These two literatures rarely intersect with churn-prediction research.

In summary, each of these research areas has developed rigorously within its own scope, but limited work has combined them to examine directly whether green purchasing behaviour is associated with customer retention, applying comparable methodological rigor across all four areas simultaneously. Table 3.1 summarises the key works reviewed above, alongside the specific limitation each leaves unresolved.

Table 3.1: Literature Review Table Showing the Contributions of Key Works

| **Year** | **Author(s)** | **Title** | **Method(s)** | **Results** | **Contribution(s)** | **Drawback/ Limitations** |
|---|---|---|---|---|---|---|
| 2024 | Ekene et al. (Ekene et al. 2024) | Predictive Analytics in E-commerce Retention | Literature review | Predictive analytics aids at-risk customer ID | Synthesizes retention analytics | E-commerce only; no leakage/sustainability angle |

Continued on next page

**Table 3.1 continued**

| Year | Author(s) | Title | Method(s) | Results | Contribution(s) | Drawback/ Limitations |
|---|---|---|---|---|---|---|
| 2025 | Rogers & Cascone (Rogers & Cascone 2025) | Sustainability Stops Being a Side Hustle | Consumer survey/trend analysis | Green-demand growth is plateauing | Shows market ceiling for green products | No empirical green-loyalty test |
| 2024 | de Lange et al. (de Lange et al. 2024) | EU Sustainability Regs. Reshaping Markets | Regulatory analysis | EU rules reshape claims/reporting | Frames regulation as strategy | No retention evidence |
| 2020 | ElHaffar et al. (ElHaffar et al. 2020) | Closing the Attitude-Behavior Gap | Narrative review (58 studies) | Confirms persistent green gap | Synthesizes green-gap theory | Attitudinal; no churn data |
| 2026 | Vo et al. (Vo et al. 2026) | Reevaluating Churn Labeling in E-Commerce | Static vs. dynamic label comparison | Dynamic labels outperform static ones | Shows label design affects outcomes | Narrow domain; conference paper |
| 2025 | Suh (Suh 2025) | Segments via Interaction Behaviors | Behavioral clustering | Interpretable segments emerge | Extends RFM to interaction data | Domain-specific; no green angle |
| 2005 | Fader et al. (Fader et al. 2005) | RFM and CLV | Iso-value curve modeling | R,F jointly predict CLV | Links RFM to CLV formally | No sustainability treatment |
| 2025 | Ling & Tang (Ling & Weiling 2025) | Comparative Study of Clustering Methods | Algorithm benchmarking | Identifies method strengths/weaknesses | Guides algorithm selection | Not applied to green/churn context |
| 2020 | Shahapure & Nicholas (Shahapure & Nicholas 2020) | Cluster Quality via Silhouette Score | Silhouette evaluation | Shows utility/limits of silhouette | Guidance on score interpretation | Purely methodological |
| 2021 | Shwartz-Ziv & Armon (Shwartz-Ziv & Armon 2021) | Tabular Data: DL Is Not All You Need | DL vs. tree-ensemble benchmark | Trees often beat DL on tabular data | Justifies tree ensembles for tabular tasks | Not churn-specific |
| 2017 | Lundberg & Lee (Lundberg & Lee 2017) | Unified Approach to Interpreting Predictions | SHAP (Shapley attribution) | Contributions sum to prediction gap | Enabled consistent explainability | Rarely tests confounding vs. real effect |

Continued on next page

**Table 3.1 continued**

| Year | Author(s) | Title | Method(s) | Results | Contribution(s) | Drawback/ Limitations |
|---|---|---|---|---|---|---|
| 2025 | Gupta et al. (Gupta et al. 2025) | Mitigation Strategies for Data Leakage | Review & mitigation framework | Proposes leakage-prevention strategies | System-level leakage framework | Not applied to churn labeling |
| 2017 | Clark & Tilman (Clark & Tilman 2017) | Environmental Impacts of Food Choice | Comparative meta-analysis | Confirms footprint variation by food type | Reinforces food-choice impact evidence | No consumer loyalty angle |
| 2020 | Adrita & Mohiuddin (Adrita & Mohiuddin 2020) | Opportunity/Ability in Green Behavior | Survey-based study | Opportunity/ability moderate attitude-behavior link | Extends green-gap moderators | Attitudinal; no transactional data |
| 2025 | Huang (Huang 2025) | Customer Segment Model via RFM | RFM-based segmentation | RFM distinguishes segments effectively | Reinforces classical RFM approach | No sustainability dimension |
| 2023 | Kasem et al. (Kasem et al. 2023) | AI Profiling & Sales Prediction | AI profiling/segmentation-/prediction | Improves profiling and sales prediction accuracy | Links segmentation to sales prediction | No leakage audit/green angle |
| 2022 | Alamsyah et al. (Alamsyah et al. 2022) | RFM + K-Means Segmentation | RFM + K-Means clustering | Produces interpretable segments | Validates RFM + K-Means pipeline | No sustainability/leakage angle |
| 2020 | Jain et al. (Jain et al. 2020) | Churn Prediction: LR and LogitBoost | Logistic regression, LogitBoost | Boosting outperforms plain LR | Benchmarks boosting vs. linear baseline | No leakage audit |
| 2011 | Kaufman et al. (Kaufman et al. 2011) | Leakage in Data Mining | Conceptual framework, case studies | Identifies recurring leakage patterns | Founding taxonomy for data leakage | Rarely audited explicitly in applied work |

## 3.7 Our Contribution

### 3.7.1 Gap Analysis

Three specific gaps, drawn directly from Table 3.1, motivate this thesis.

**Gap 1 – Segmentation frameworks omit sustainability behaviour.** RFM-based segmentation, however refined, has no established place for a customer's green-spending pattern (Fader et al. 2005). A segmentation exercise conducted using only

classical RFM inputs would not surface a persona such as "young, high green-share, low-spending," not because this persona is absent from the data, but because the framework was not designed to identify it.

**Gap 2 – Churn studies rarely audit their own labels.** Despite leakage being a well-established risk in principle (Kaufman et al. 2011), published churn case studies (Jain et al. 2020) typically report performance metrics without demonstrating that the label itself was examined for entanglement with the model's own features. This leaves open the possibility that some reported churn-model accuracy in the literature is, at least in part, an artefact of label construction rather than a genuine predictive achievement.

**Gap 3 – The green-loyalty assumption is asserted, not tested.** The proposition that sustainability-minded customers are also more loyal appears frequently in industry commentary (de Lange et al. 2024, Rogers & Cascone 2025), yet the academic literature on the green gap (ElHaffar et al. 2020) points toward the opposite conclusion, expressing scepticism that stated or even actual green behaviour reliably predicts other outcomes. The literature reviewed in this chapter provided no example of this specific claim being tested directly against real transactional churn data using a model whose own label had been audited for leakage. This thesis does not claim to have conducted an exhaustive systematic review; it remains possible that such a study exists outside the sources consulted here, but none was identified within the scope of the literature search this chapter describes.

### 3.7.2 Novelty of This Study

The novelty of this thesis does not lie in any single technique: RFM, K-Means, XGBoost, and SHAP are all well-established methods, and this thesis does not claim to extend any of them individually. The novelty lies in their combination, and specifically in a diagnostic step without clear precedent in the reviewed literature: identifying, demonstrating, and correcting data leakage in a churn label constructed for this purpose, using a real customer dataset, and carrying the corrected model through segmentation and SHAP-based explanation. The use of SHAP in this thesis is similarly distinct: rather than reporting feature importance alone, SHAP is used specifically to examine whether an apparent green-spending effect is genuine or reflects a proxy relationship, an application not identified elsewhere in the sustainability-marketing literature reviewed.

### 3.7.3 Significance of Our Work

The significance of this thesis is primarily practical. If the assumption that green customers are more loyal does not hold, as this thesis's findings suggest, this carries

a direct implication for marketing resource allocation: green initiatives and retention initiatives should be planned, resourced, and evaluated as separate efforts, rather than assumed to reinforce one another automatically. Beyond this specific finding, this thesis offers a methodological template, leakage-aware label construction combined with segmentation and explainability, that could reasonably be reused by researchers or analysts seeking to test, rather than assume, a comparable business hypothesis.

# Chapter 4 Methodology

## 4.1 Dataset

The dataset used in this study was the marketing_campaign dataset, obtained from Kaggle. It comprised a single tab-separated file containing 2,240 customer records and 29 raw variables. These variables were conceptually organised into four groups: demographic data (birth year, education, marital status, income, number of children and teenagers at home), product spending (spending across six categories, wine, fruits, meat, fish, sweets, and gold/premium products), promotion history (acceptance of five previous marketing campaigns and response to the most recent campaign), and channel data (number of purchases through deals, web, catalogue, and physical store, plus monthly website visits), together with enrolment date and recency (days since the last purchase).

This combination of variables made the dataset suitable for the present study: the purchase-channel and spending variables supported the construction of RFM (Recency, Frequency, Monetary) features and the inactivity label, while the category-level spending data enabled construction of a green-behaviour proxy, allowing the relationship between sustainable purchasing and inactivity to be examined directly. Two additional columns present in the raw file, Z_CostContact and Z_Revenue, held a constant value across all customer records and carried no analytical information; these were excluded from Table 4.1 and from all subsequent analysis. Table 4.1 provides an overview of the remaining 27 variables' structure prior to processing.

## 4.2 Data Pre-Processing

Data cleaning proceeded through the following steps. First, 24 rows with missing income values were removed, since income could not be reliably imputed and represented a small fraction of the dataset. Second, the enrolment date (Dt_Customer) was parsed into datetime format, and the most recent enrolment date across all customers was recorded as a reference date. This reference date served two pur-

poses: computing each customer's age from their birth year, and computing each customer's tenure in days (Days_As_Customer) as the interval between their enrolment date and the reference date, since raw birth years and date strings are not directly usable as numeric features. Third, implausible records were removed as outliers: three customers with an implied age above 90 years (corresponding to birth years such as 1893) and one customer with income exceeding EUR 200,000, identified as data-entry errors that would distort both clustering distances and summary statistics. Fourth, rare or non-sensical categories in the Marital_Status field (Absurd, Alone, Widow, YOLO) were consolidated into Single. Fifth, a Total_Kids feature was derived as the sum of children and teenagers in the household. Following cleaning, 2,212 customer records remained (98.8% of the original dataset), forming the sample used throughout the remainder of the analysis.

Table 4.1: Overview of the *marketing_ campaign* Dataset Structure (Before Processing)

| **Category** | **Variable** | **Description** |
|---|---|---|
| People | ID | Customer's unique identifier |
| | Year_Birth | Customer's birth year |
| | Education | Customer's education level |
| | Marital_Status | Customer's marital status |
| | Income | Customer's yearly household income |
| | Kidhome | Number of children in customer's household |
| | Teenhome | Number of teenagers in customer's household |
| | Dt_Customer | Date of customer's enrollment with the company |
| | Recency | Number of days since customer's last purchase |
| | Complain | 1 if the customer complained in the last 2 years, 0 otherwise |
| Products | MntWines | Amount spent on wine in the last 2 years |
| | MntFruits | Amount spent on fruits in the last 2 years |
| | MntMeatProducts | Amount spent on meat in the last 2 years |
| | MntFishProducts | Amount spent on fish in the last 2 years |
| | MntSweetProducts | Amount spent on sweets in the last 2 years |
| | MntGoldProds | Amount spent on gold in the last 2 years |
| Promotion | NumDealsPurchases | Number of purchases made with a discount |
| | AcceptedCmp1 | 1 if customer accepted the offer in the 1st campaign, 0 otherwise |
| | AcceptedCmp2 | 1 if customer accepted the offer in the 2nd campaign, 0 otherwise |

Continued on next page

**Table 4.1 continued from previous page**

| Category | Variable | Description |
|---|---|---|
| | AcceptedCmp3 | 1 if customer accepted the offer in the 3rd campaign, 0 otherwise |
| | AcceptedCmp4 | 1 if customer accepted the offer in the 4th campaign, 0 otherwise |
| | AcceptedCmp5 | 1 if customer accepted the offer in the 5th campaign, 0 otherwise |
| | Response | 1 if customer accepted the offer in the last campaign, 0 otherwise |
| Place | NumWebPurchases | Number of purchases made through the company's website |
| | NumCatalogPurchases | Number of purchases made using a catalogue |
| | NumStorePurchases | Number of purchases made directly in stores |
| | NumWebVisitsMonth | Number of visits to the company's website in the last month |

## 4.3 Feature Engineering

### 4.3.1 RFM features (Frequency and Monetary)

RFM (Recency, Frequency, Monetary) is a well-established method for summarising a customer's transactional value in three quantities. As Recency already existed in the raw data, only Frequency and Monetary required computation. Monetary was constructed as the sum of spending across all six product categories (wine, fruits, meat, fish, sweets, and gold/premium products).

Frequency was initially constructed by summing purchases across all four available channel variables, including NumDealsPurchases (discount-driven purchases). On closer examination, this risked double-counting, since a discounted purchase is not necessarily a separate channel from web, catalogue, or store purchases; including it alongside those three channels could therefore count the same transaction twice. Three diagnostic checks were conducted to assess this concern. First, NumDealsPurchases exceeded the sum of the other three channels for only 3 of 2,212 customers (0.1%), consistent with deals purchases occurring through, rather than separately from, the three channels. Second, the correlation between NumDealsPurchases and total channel purchases was weak ($r = 0.117$), suggesting deals usage does not scale consistently with overall purchase volume. Third, 98.2% of customers with any

channel purchases also recorded at least one deals purchase (40 of 2,212 customers, 1.8%, had zero deals purchases but nonzero channel purchases). This evidence was suggestive rather than conclusive; the more conservative choice was nonetheless to exclude NumDealsPurchases from Frequency to eliminate any risk of double-counting. Frequency was therefore redefined as the sum of purchases through the three unambiguous channels: web, catalogue, and store. NumDealsPurchases was retained separately as an independent behavioural variable. Both Frequency and Monetary served as candidate inputs for clustering, and Monetary was additionally required as an input to the green-spending ratio described below.

### 4.3.2 Green_Score and Green_Ratio

Green_Score was computed as the sum of spending on fruit and fish products. Meat spending was deliberately excluded from this measure, since meat production carries a disproportionately higher environmental footprint than fruit or fish production; its inclusion in a "green" measure would have worked against the purpose the feature was intended to capture. Green_Ratio was computed as Green_Score divided by Monetary, converting an absolute spending amount into a proportion of total spending bounded between 0 and 1. An earlier version of this formula divided by (Monetary + 1), a common safeguard against division by zero; this was removed once it was confirmed that no customer in the cleaned sample had zero Monetary spending (minimum EUR 5.00), since the added constant was unnecessary and introduced a slight distortion for low-spending customers. This ratio was used, rather than a raw amount, because it captures a customer's green *preference*, the proportion of total spending directed toward these lower-footprint categories, rather than overall spending level; a high-spending and a low-spending customer could show an equivalent proportional preference despite substantially different raw green-spending amounts. Green_Ratio should be understood as a spending-based proxy rather than a confirmed measure of environmental motivation: the dataset does not indicate whether the fruit or fish purchased was sustainably sourced, and spending amount is not equivalent to environmental impact, since price differences between products mean equal spending does not imply equal purchased quantity or footprint. This proxy is treated accordingly throughout the remainder of this thesis, and the term *green-spending proxy* is used in preference to terms such as *eco-conscious* or *sustainability-motivated* when describing customers or segments.

### 4.3.3 Total_Accepted_Campaigns

This feature was constructed as the sum across the five historical campaign-acceptance columns, providing a single measure of each customer's overall marketing engage-

ment.

### 4.3.4 Descriptive Bucketing (Green_Level, Income_Level)

Categorical versions of Green_Ratio (Low, Medium, and High Green, using thresholds of 0.10 and 0.25) and Income (four quartiles: Low, Medium-Low, Medium-High, High) were also constructed. Continuous ratios and euro amounts are difficult to compare directly across groups, so bucketing them into discrete categories enabled comparison of inactivity rates and spending across clearly defined groups during exploratory analysis. These buckets were used solely for descriptive purposes and were not included as model inputs.

### 4.3.5 Churn Label Construction and Data-Leakage Correction

An initial candidate label defined a customer as churned if their Recency exceeded 90 days, or if they had accepted zero marketing campaigns. This formulation initially appeared reasonable, since both prolonged purchase inactivity and total disengagement from marketing campaigns constitute plausible indicators of disloyalty. On inspection, however, this formulation was found to introduce data leakage: because Total_Accepted_Campaigns was also intended for use as a predictive feature, and because the majority of customers (1,754 of 2,212, 79.3%) had accepted zero campaigns, a model trained on this label could largely learn the rule that zero campaigns accepted implies churn, achieving inflated performance not by identifying a genuine behavioural pattern, but by partially reconstructing the rule used to define its own label. To eliminate this leakage, the label used throughout the remainder of this thesis was redefined using a single signal:

$$\text{Churn} = \begin{cases} 1 & \text{if Recency} > 90 \text{ days} \\ 0 & \text{otherwise} \end{cases}$$

This single-signal definition avoids leakage because Recency was excluded entirely from the set of model features, so no information from the label's construction could re-enter the model as a predictor.

The choice of 90 days as the inactivity threshold also warrants justification, since Recency in the cleaned sample ranges only from 0 to 99 days; a 90-day cutoff selects customers from the extreme upper tail of this distribution, and the resulting inactivity rate is consequently sensitive to the threshold chosen. This threshold was adopted as a round, interpretable value in the absence of a domain-specific definition

of inactivity for this retailer. To confirm that the study's central findings do not depend on this specific choice, a sensitivity analysis was conducted at two additional thresholds (60 and 75 days), reported in Section 4.3.6.

### 4.3.6 Threshold Sensitivity Analysis

To confirm that the study's central findings do not depend on the specific 90-day threshold, the inactivity label and its correlation with Green_Ratio were recomputed at two additional thresholds (60 and 75 days), shown in Table 4.2.

Table 4.2: Threshold Sensitivity Analysis for the Inactivity Proxy

| **Threshold** | **Customers Classified Inactive** | **Percentage** | **Correlation with Green_Ratio** |
|---|---|---|---|
| Recency $>$ 60 days | 849 | 38.4% | $-0.0368$ |
| Recency $>$ 75 days | 524 | 23.7% | $-0.0030$ |
| Recency $>$ 90 days | 194 | 8.8% | $-0.0045$ |

The correlation between Green_Ratio and inactivity status remains negligible across all three thresholds, ranging from $-0.0368$ at the 60-day threshold to $-0.0030$ at the 75-day threshold. Although the magnitude at the 60-day threshold exceeds that at 75 or 90 days, all three values remain well below conventional thresholds for a meaningful linear association (typically $|r| > 0.1$), indicating that the study's central finding regarding the absence of a green-spending/inactivity relationship does not depend substantially on the specific 90-day cutoff chosen.

### 4.3.7 Terminology: Churn as an Inactivity Proxy

Because Recency deterministically determines this label, the model described in Section 4.7 does not predict future churn; it examines whether other customer characteristics are associated with this inactivity status. Once a customer's Recency value is known, their classification under this definition is known with certainty, so the task addressed by the model is not forecasting a future event, but assessing association between baseline customer characteristics and a present-moment inactivity status. Accordingly, the term *90-day inactivity proxy* is used throughout the remainder of this thesis in preference to *future churn*, except where referring informally to the underlying business concern motivating the analysis. A genuine churn-prediction design would require predictors measured at time $t$ and an outcome observed in a

later period $t+1$; the cross-sectional nature of the available dataset does not permit this design, a limitation discussed further in Chapter 6.

## 4.4 EDA (Exploratory Data Analysis)

### 4.4.1 Descriptive Statistics of Key Features

Summary statistics (mean, standard deviation, minimum, median, and maximum) were computed across the key numerical features, Age, Income, Recency, Frequency, Monetary, Green_Ratio, Total_Kids, Days_As_Customer, and the inactivity label. This established a baseline understanding of the customer base's typical profile and value ranges prior to grouping or modelling, and served as a sanity check that the cleaning and feature-engineering steps had produced sensible values.

### 4.4.2 Business Question 1: Overall Inactivity Rate

Using the leakage-free label, the overall proportion of customers classified as inactive was computed. This is the most fundamental quantity the analysis depends on, and was accordingly established descriptively before any predictive model was asked to reproduce or explain it.

### 4.4.3 Business Question 2: Do Green Customers Churn Less?

The Green_Level buckets were used to compare inactivity rates across Low, Medium, and High Green customer groups, and the correlation between Green_Ratio and inactivity was separately computed across the full sample. This provided a direct descriptive test of whether green spending relates to inactivity, prior to any predictive model, so that later model-based results could be assessed against this baseline.

### 4.4.4 Business Question 3: Income, Churn, and Green Spending

Income_Level quartiles were used to examine whether income relates to either green spending or inactivity rate. Income was treated as a candidate confounding variable at this early stage: if income alone explained both green spending and inactivity, this would suggest that any apparent green-inactivity relationship might reflect a proxy relationship with income, a consideration directly relevant to the interpretation of the SHAP results in Chapter 5.

### 4.4.5 Business Question 4: Correlation Analysis

A correlation matrix was computed across Recency, Frequency, Monetary, Green_Ratio, Income, Age, Total_Kids, Days_As_Customer, Total_Accepted_Campaigns, and the inactivity label, and visualised as a heatmap. This identified, in linear terms, which variables showed any association with the label before a more complex model was introduced. It also permits a methodological observation: because Recency defines the label by construction, it is expected, and correct, that no other feature shows strong correlation with the label once Recency is excluded as a predictor.

### 4.4.6 Distribution Plots

The distributions of the key numerical features (Age, Income, Recency, Frequency, Monetary, Green_Ratio) were visualised as histograms. Assessing whether features were skewed, bounded, or approximately normal was relevant to later decisions, particularly the justification for standardising features prior to K-Means clustering (Section 4.6.1), since several of these features are right-skewed and measured on markedly different scales.

## 4.5 Statistical Testing of the Green-Spending and Inactivity Relationship

The descriptive comparison in Business Question 2 and the correlation reported in Business Question 4 provide useful initial indications, but neither constitutes a formal significance test, and a simple linear correlation cannot rule out a non-linear relationship. Three complementary tests were used to examine the green-spending/inactivity relationship more rigorously.

First, a chi-square test of independence was applied to the contingency table of Green_Level against the inactivity label, together with Cramér's V as a measure of effect size, to test whether observed differences in inactivity rate across green-spending groups exceeded what would be expected by chance. Ninety-five percent confidence intervals (Wilson score method) were also computed around each group's inactivity rate, to characterise the degree of overlap between groups directly.

Second, a logistic regression of the inactivity label on continuous Green_Ratio was fit, to test for a linear association without collapsing Green_Ratio into discrete bins.

Third, since a linear model cannot capture a relationship that rises and falls non-

monotonically, a quadratic term (Green_Ratio$^2$) was added to the logistic regression, and a likelihood-ratio test was used to assess whether the quadratic model fit significantly better than the linear model. Given the limited number of positive cases (194 across the full sample), this nonlinearity check was kept deliberately simple, using a single squared term rather than a more flexible spline, to avoid overfitting a small number of positive examples.

The results of all three tests are reported in Chapter 5 alongside the descriptive comparison, and are revisited in Section 4.5.1.

### 4.5.1 Multivariate Logistic Regression

The SHAP analysis described in Section 4.8 shows the model assigning some importance to Green_Ratio despite the near-zero raw correlation reported above. One plausible explanation is confounding: Green_Ratio may be statistically entangled with other variables, such as Income or Age, that themselves relate to inactivity, causing the model to rely on Green_Ratio as an indirect proxy. This possibility cannot be assessed from a SHAP dependence plot alone, since a confounder must be shown to relate to *both* the feature in question and the outcome, not merely to co-vary with the feature's plotted colour scale. A multivariate logistic regression was therefore fit, predicting the inactivity label from Green_Ratio together with Income, Age, Total_Kids, and Days_As_Customer as controls, all standardised to a common scale. If Green_Ratio remained non-significant after these variables were accounted for, and if the control variables themselves showed no significant relationship with inactivity, this would argue against a straightforward confounding explanation. The results of this test are reported and interpreted in Chapter 5.

## 4.6 Customer Segmentation via K-means Clustering

### 4.6.1 Feature preparation for clustering

Eight features, Income, Age, Recency, Frequency, Monetary, Green_Ratio, Total_Kids, and Days_As_Customer, were used for clustering. These features were standardised (zero mean, unit variance) using StandardScaler prior to clustering. This step was necessary because K-Means measures similarity using Euclidean distance, so features on larger absolute scales, such as Income, measured in tens of thousands, would otherwise dominate the distance calculation and effectively diminish the influence of features on smaller, bounded scales, such as Green_Ratio,

despite their comparable analytical importance. Standardisation placed all features on equal footing prior to distance calculation.

### 4.6.2 Selecting the Number of Clusters (K)

K-Means was fit for a range of $k$ values (2 through 10), and two diagnostics were examined jointly: inertia (within-cluster sum of squared distances, via the elbow method) and the silhouette score. The elbow method identifies the point at which additional clusters cease to meaningfully reduce inertia, while the silhouette score quantifies how well-separated and internally cohesive the resulting clusters are for a given $k$.

The silhouette score was highest at $k = 2$, but this value was deliberately rejected, since it separates customers only into a coarse high-value/low-value split, insufficiently granular to support differentiated marketing action. A final value of $k = 5$ was selected instead, explicitly trading statistical cluster cohesion for a segmentation of greater practical usefulness to a business audience. This choice is made explicit and justified quantitatively in this thesis, rather than presented as a purely data-driven optimum.

### 4.6.3 Applying K-Means and Building Segment Profiles

K-Means was fit with $k = 5$ on the standardised features, assigning each customer to one of five clusters. The raw numerical profile of each cluster (average Age, Income, Recency, Frequency, Monetary, Green_Ratio, Total_Kids, Days_As_Customer, and inactivity rate) was inspected before segment names were assigned. This ordering was deliberate: naming clusters prior to inspection risks projecting an assumed narrative onto the data, whereas inspecting the raw profiles first ensures that segment names describe what was actually observed. Descriptive, business-facing names were assigned only after this inspection, based on each cluster's distinguishing characteristics. Supporting visualisations were also produced, including segment size distribution, inactivity rate by segment, average green ratio by segment, and an income-versus-spend scatter plot coloured by segment.

### 4.6.4 Cluster Stability

Because K-Means initialises centroids randomly, resulting cluster assignments can vary between runs. To assess the stability of the five-segment solution, K-Means was rerun five additional times with different random seeds, and each run's cluster assignments were compared to the original solution using the Adjusted Rand Index

(ARI), where 1.0 indicates identical assignments and 0.0 indicates agreement no better than chance. Agreement was moderate (mean ARI = 0.767, range 0.551–1.000 across seeds), indicating that the clustering captures genuine structure in the data, while individual customer-to-segment assignments remain somewhat sensitive to initialisation; the five-segment solution should accordingly be interpreted as one reasonable partition of the data rather than a uniquely determined one. Two additional validation metrics were computed for context: a Davies-Bouldin Index of 2.009 and a Calinski-Harabasz score of 480.08.

As a further check, a Gaussian Mixture Model (GMM) was fit to the same standardised features. Agreement with the K-Means solution was low (ARI = 0.115), and the resulting GMM clusters were highly imbalanced in size (ranging from 45 to 978 customers), in contrast to the comparatively even K-Means segments (380–501 customers). This divergence most plausibly reflects GMM's assumption of elliptical, Gaussian-shaped clusters, which may not suit this dataset's structure, rather than evidence that the K-Means solution itself is unreliable; it nonetheless indicates that the five-segment structure is specific to K-Means and should not be assumed to generalise automatically to other clustering approaches.

### 4.6.5 Sensitivity to Feature Skewness

Several clustering inputs, particularly Monetary and Green_Ratio, are right-skewed (Section 5.1.1). Although standardisation places all features on a common scale, it does not correct skewness, and K-Means, which relies on Euclidean distance, remains susceptible to disproportionate influence from extreme values. To examine this, clustering was repeated using a log-transformed version of Monetary. The log-transformed version achieved a marginally higher silhouette score (0.162 versus 0.150), but agreement with the original cluster assignments was low (ARI = 0.361), indicating that the skewness of Monetary meaningfully influences cluster membership. Given the disruption a full re-segmentation would introduce to the remainder of the analysis, and the modest magnitude of the silhouette improvement, the original clustering was retained as the primary segmentation for this thesis; this sensitivity is reported as a limitation, discussed further in Chapter 6.

## 4.7 Churn Prediction Model

### 4.7.1 Feature Selection for Churn Prediction

Following the leakage correction described in Section 4.3.5, an initial, conservative feature set of five variables was used: Income, Age, Green_Ratio, Total_Kids, and Days_As_Customer. Recency was excluded because it defines the label by construction (Churn = Recency $>$ 90), and its inclusion would render the prediction task trivial and circular. Frequency, Monetary, and Total_Accepted_Campaigns were initially excluded as well, on the basis that they had been entangled in the earlier, rejected label formulation.

This initial exclusion conflated two distinct concerns: direct leakage, where a variable definitionally determines the label, and correlation with a variable used in an earlier, discarded label formulation. Only Recency satisfies the first criterion; Frequency, Monetary, and Total_Accepted_Campaigns do not, since the final label depends on Recency alone. Table 4.3 makes this distinction explicit by classifying each candidate feature according to its availability relative to the outcome.

Table 4.3: Feature Availability Relative to the Inactivity Label

| **Variable** | **Measurement Window** | **Available Before Outcome?** |
|---|---|---|
| Income | Baseline | Yes |
| Age | Baseline | Yes |
| Total_Kids | Baseline | Yes |
| Days_As_Customer | Baseline | Yes |
| Green_Ratio | Prior period | Yes |
| Frequency | Prior period | Yes |
| Monetary | Prior period | Yes |
| Total_Accepted_Campaigns | Prior period | Yes |
| Recency | Defines outcome | No (direct leakage) |

To test whether the original, conservative exclusion had suppressed genuine predictive signal, an extended eight-feature set was constructed, reintroducing Frequency, Monetary, and Total_Accepted_Campaigns alongside the original five variables. Both the five-feature and eight-feature model versions are reported in Chapter 5, together with a paired bootstrap comparison assessing whether the extended set produced a statistically meaningful improvement.

### 4.7.2 Train/Test Split

The data were split into training (75%, $n$ = 1,659) and test (25%, $n$ = 553) sets using stratified sampling. Because the inactivity rate is low (8.8%), a plain random split risked producing a test set with a meaningfully different proportion than the training set by chance; stratification preserved the same proportion in both sets, improving the reliability of the resulting evaluation.

### 4.7.3 Benchmark Model: Logistic Regression

A logistic regression model was trained as a baseline, using `class_weight="balanced"` to compensate for class imbalance, on standardised features. A baseline model provides a simple, interpretable point of reference, allowing the performance of a more complex model to be assessed against a known standard rather than reported in isolation.

### 4.7.4 Main Model: XGBoost

XGBoost was selected as the primary model, a widely used algorithm for structured, tabular data, capable of capturing non-linear relationships and feature interactions that logistic regression cannot represent. Class imbalance was addressed through the `scale_pos_weight` parameter, set to the ratio of negative to positive training examples (10.36), which instructs the model to weight errors on the minority class more heavily.

### 4.7.5 Hyperparameter Tuning

XGBoost's hyperparameters (`n_estimators`, `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`) were tuned using RandomizedSearchCV, sampling 30 configurations and scoring each using 5-fold stratified cross-validation on the training set. Randomised search was selected over exhaustive grid search for its efficient coverage of the hyperparameter space, avoiding the combinatorial cost of testing every parameter combination.

## 4.8 Model Explainability with SHAP

Because XGBoost, like other tree-ensemble models, does not offer directly interpretable outputs in the manner of logistic regression coefficients, an explainability step was required to examine what the trained model relied upon. This was particularly relevant given that a central research question of this thesis concerns whether

green spending is genuinely associated with inactivity, rather than simply whether inactivity can be predicted. SHAP (SHapley Additive exPlanations) was used for this purpose: a method that assigns each feature a contribution value for every individual prediction, based on Shapley values from cooperative game theory (Lundberg & Lee 2017). Specifically, `shap.TreeExplainer` was used with its default configuration, as it is optimised for tree-based ensemble models such as XGBoost, computing exact SHAP values efficiently rather than relying on slower, model-agnostic approximations. SHAP values are reported in the model's raw margin (log-odds) output space, consistent with TreeExplainer's default behaviour for XGBoost classifiers; the summary plots, dependence plots, and waterfall plot presented in Chapter 5 should be interpreted in this space rather than as direct probability contributions.

It is stated explicitly, prior to presenting these results, what SHAP can and cannot establish: SHAP explains the behaviour of the fitted model, describing which features that specific model relied upon and in which direction, but it does not validate that the model's relationships are correct, causal, or generalisable beyond the model itself. This distinction is particularly relevant here, since the model's discriminative performance (Chapter 5) is only marginally better than chance; a SHAP importance ranking produced by a model of this performance should accordingly be interpreted as exploratory rather than confirmatory. SHAP values were computed for all 553 test-set customers, rather than a smaller sample, since the complete held-out set produces more statistically reliable global importance estimates than a subsample would.

A global feature-importance bar chart was produced, ranking features by their mean absolute SHAP value across the test set, providing a high-level indication of which features the model relies upon most.

A beeswarm summary plot was also produced, showing the magnitude and direction of each feature's effect for every individual test customer. This complements the bar chart, which indicates only average magnitude, by revealing the direction of each feature's influence and its consistency across customers.

Dependence plots were produced for Green_Ratio and Days_As_Customer, each coloured by Income. Given that income was identified as a candidate confounding variable during exploratory analysis, these plots were designed specifically to examine whether the model's apparent reliance on Green_Ratio or tenure reflected underlying income differences, rather than an independent effect of green spending or tenure.

Mean absolute SHAP values were also computed separately within each of the five customer segments identified during clustering, since a single global importance ranking can obscure differences in the factors driving predictions for different customer personas; this breakdown was designed to surface segment-specific variation in feature reliance.

Finally, a waterfall plot was produced for an individually selected test customer, illustrating how each feature contributed to that specific prediction relative to the model's average output. This provides a granular illustration of how the model's global logic applies to an individual case, serving both as a sanity check on the model's reasoning and as an illustrative example for a non-technical audience.

Collectively, these explainability outputs were used to determine whether the model's reliance on Green_Ratio reflected a genuine relationship with inactivity, or was better explained by confounding with other features, a question addressed directly in Chapter 5.

## 4.9 Evaluation Criterion

### 4.9.1 Why AUC, Not Accuracy

ROC-AUC was selected as the primary evaluation metric rather than raw accuracy. Because inactivity is rare in this dataset (8.8%), a model could achieve high accuracy (approximately 91%) by classifying every customer as not inactive, without learning anything substantive. AUC, by contrast, measures how well a model ranks inactive customers above non-inactive customers across all possible decision thresholds, providing a considerably more informative measure under severe class imbalance.

### 4.9.2 Supporting Metrics

Alongside AUC, confusion matrices and full classification reports (precision, recall, and F1-score for both classes) were produced. AUC is a single aggregate value and does not indicate performance at any specific decision threshold; precision and recall for the inactive class specifically indicate whether the model's classifications would be useful in practice, that is, of the customers flagged as inactive, how many genuinely are, and of the customers who are genuinely inactive, how many are correctly identified.

### 4.9.3 Train/Test Evaluation Strategy

All reported test-set metrics were computed on the stratified 25% held-out test set described in Section 4.7.2, which played no role in model training or hyperparameter tuning. This distinction is relevant because, as discussed below, a separate AUC estimate was also produced during tuning, and the two are not directly comparable.

### 4.9.4 The Cross-Validation vs Held-Out Test Set Distinction

Two distinct AUC estimates exist in this thesis: one obtained from 5-fold cross-validation during hyperparameter tuning, and one obtained from the held-out test set, which played no role in tuning. Of the 194 inactive customers in the dataset, 146 fall within the training set, leaving approximately 29 positive examples per cross-validation fold; this limited per-fold count renders the cross-validation AUC estimate noisy, and the hyperparameters selected through this process were consequently determined partly by this noise rather than by fully generalisable signal. The held-out test set, by contrast, was not accessed during tuning, so its AUC provides an independent estimate of performance, though one that retains considerable sampling uncertainty given the limited number of positive test cases (48). Accordingly, the test-set AUC is treated throughout this thesis as the primary reportable measure of performance, while the cross-validation AUC is treated as an optimistic, tuning-time estimate rather than as the model's expected real-world performance. Bootstrap confidence intervals, described below, were used to quantify this uncertainty directly.

### 4.9.5 Bootstrap Confidence Intervals for AUC

Given the limited number of positive test cases, a single test-set AUC value can be misleading in isolation. To address this, 95% confidence intervals were constructed for each model's AUC using bootstrap resampling (1,000 iterations, resampling the test set with replacement). For comparisons between two models, a paired bootstrap was used instead, resampling identical indices for both models on each iteration and computing the distribution of their AUC difference; this reflects the fact that both models were evaluated on the same customers, producing a tighter and more appropriate estimate of the true performance gap than would be obtained by comparing two independently bootstrapped intervals.

# Chapter 5 Results and Discussions

## 5.1 Results

### 5.1.1 Exploratory Data Analysis

Descriptive statistics for the key numerical features are reported in Table 5.1. The average customer is 45.1 years old, earns approximately EUR 51,959, last purchased 49.0 days prior on average, made 14.9 purchases in total, spent EUR 607.27 in total, and has a Green_Ratio of 0.12.

textbfBusiness Question 1: What is the overall inactivity rate? Under the leakage-free definition (Recency $>$ 90 days), the overall inactivity rate is 8.8%, with 194 of 2,212 customers classified as inactive.

**Business Question 2**: Do customers with higher green-spending show lower inactivity? Customers were bucketed into Low Green (1,196), Medium Green (737), and High Green (279) based on Green_Ratio. Inactivity rates across these groups are 9.4%, 8.1%, and 7.9% respectively, and the correlation between Green_Ratio and inactivity is approximately $-0.004$. This descriptive comparison alone does not establish statistical significance and was accordingly followed by the formal tests described in Section 4.5 and reported in full in Section 5.1.2.

**Business Question 3**: How does income relate to inactivity and green spending? Green_Ratio does not vary monotonically with Income_Level (Low: 0.185; Medium-Low: 0.089; Medium-High: 0.089; High: 0.117), nor does inactivity rate (8.5%, 9.9%, 6.5%, 10.1% respectively).

**Business Question 4**: Which features correlate most strongly with inactivity? Once Recency is excluded, no remaining feature correlates with the inactivity label above $|r| = 0.03$ (Age: $-0.026$; Green_Ratio: $-0.004$; Total_Accepted_Campaigns: 0.000; Income: 0.004; Monetary: 0.006; Frequency: 0.010).

### 5.1.2 Statistical Testing of the Green-Spending and Inactivity Relationship

Following the methodology described in Section 4.5, three tests were conducted to examine whether the descriptive pattern above holds under formal testing.

A chi-square test of independence across the three Green_Level groups found no statistically significant association with inactivity ($\chi^2 = 1.166$, $df = 2$, $p = 0.558$), with a negligible effect size (Cramér's V = 0.023). Ninety-five percent confidence intervals around each group's inactivity rate overlapped substantially: Low Green 9.4% [7.8%, 11.1%], Medium Green 8.1% [6.4%, 10.3%], High Green 7.9% [5.3%, 11.6%].

A simple logistic regression of the inactivity label on continuous Green_Ratio confirmed this result: the Green_Ratio coefficient was not statistically significant ($\beta = -0.152$, odds ratio = 0.859, 95% CI [0.221, 3.349], $p = 0.827$).

A nonlinearity check, adding a quadratic Green_Ratio term to the regression, similarly found no meaningful improvement in fit. The quadratic term itself was not significant ($\beta = -1.569$, $p = 0.746$), and a likelihood-ratio test comparing the linear and quadratic models found no significant difference (LR = 0.107, $p = 0.743$).

Taken together, the chi-square test, the linear logistic regression, and the nonlinearity check are in agreement: no statistically detectable relationship exists between green-spending behaviour and inactivity in this dataset, in any of the functional forms tested.

Table 5.1: Descriptive Statistics of Key Features (n = 2,212)

| **Feature** | **Mean** | **Std** | **Min** | **Median** | **Max** |
|---|---|---|---|---|---|
| Age (years) | 45.09 | 11.70 | 18.00 | 44.00 | 74.00 |
| Income (EUR) | 51,958.81 | 21,527.28 | 1,730.00 | 51,371.00 | 162,397.00 |
| Recency (days) | 49.02 | 28.94 | 0.00 | 49.00 | 99.00 |
| Frequency | 14.89 | 7.67 | 0.00 | 15.00 | 44.00 |
| Monetary (EUR) | 607.27 | 602.51 | 5.00 | 397.00 | 2,525.00 |
| Green_Ratio | 0.12 | 0.11 | 0.00 | 0.09 | 0.59 |
| Inactivity (0/1) | 0.09 | 0.28 | 0.00 | 0.00 | 1.00 |

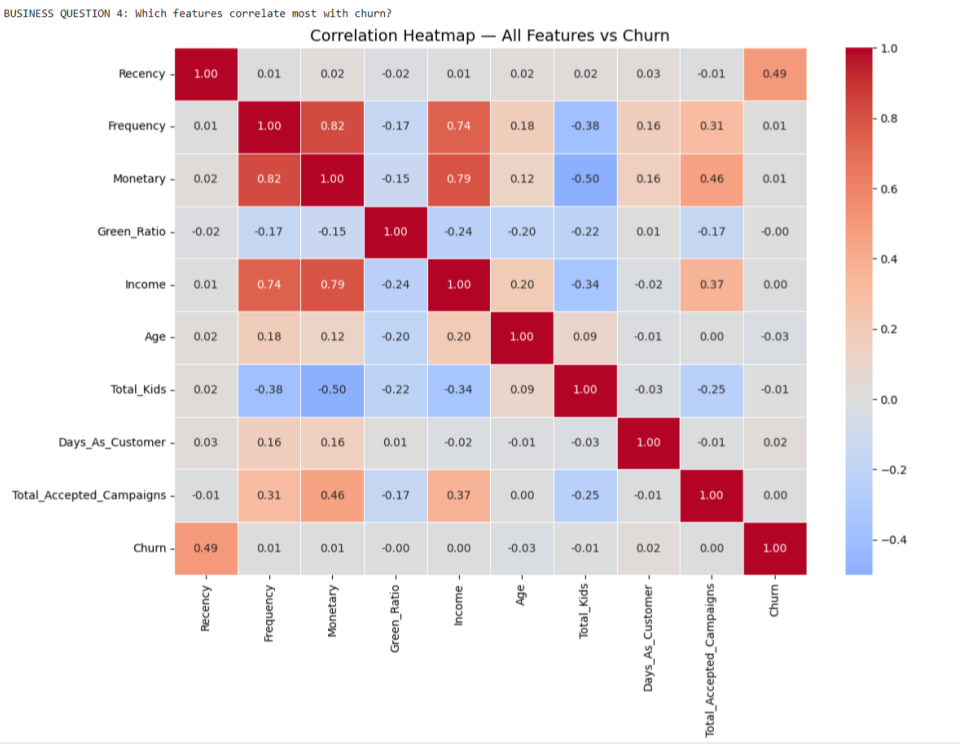



Figure 5.1: Correlation Heatmap of Engineered Features versus Inactivity

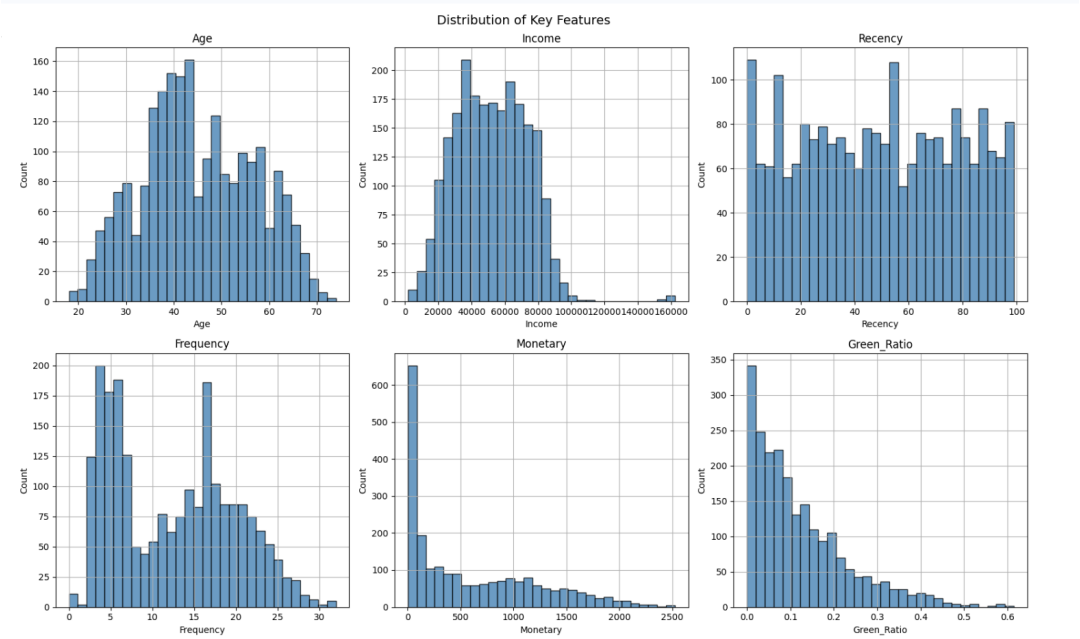



Figure 5.2: Distributions of Key Numerical Features

### 5.1.3 Customer Segmentation Results

The silhouette score was maximised at $k = 2$ (0.260) and declined thereafter, with a local plateau at $k = 5$ (0.150). K-Means with $k = 5$ produced five clusters ranging in size from 380 to 501 customers. Each cluster's raw numerical profile was

inspected prior to assigning a descriptive name; consistent with the reasoning set out in Section 5.2.1, names were kept deliberately neutral and behaviour-based, rather than attributing psychological traits, such as loyalty or environmental motivation, that the clustering variables cannot establish. Table 5.2 reports the profile and name of each segment:

1. **Budget-Constrained Family Customers (445):** most children/teenagers (1.43 average), lowest green-spending ratio (0.06), modest income and spend, 0% inactivity.

2. **Premium High-Spenders (434):** highest income (EUR 75,382) and spend (EUR 1,174), fewest children (0.36 average), green ratio 0.12, inactivity 7%.

3. **Young High Green-Proxy Customers (380):** youngest (36.6 years), lowest income (EUR 27,207), highest green-spending ratio (0.28), lowest frequency (5.91), inactivity 8%.

4. **High-Recency Low-Spend Customers (452):** highest average recency (74.91 days), highest inactivity rate (19%).

5. **High-Frequency Shoppers (501):** highest purchase frequency (20.00), solid income (EUR 67,870), inactivity 9%.

As discussed further in Section 5.2.1, these segment-level inactivity rates should be interpreted descriptively rather than as independent confirmation of the statistical tests in Section 5.1.2, since Recency is one of the eight clustering inputs and also defines the inactivity label itself.

Table 5.2: Customer Segment Profiles (K-Means, k = 5)

| Segment | Count | Avg Age | Avg Income | Avg Recency | Avg Freq. | Avg Monetary | Avg Green | Avg Kids | Inactivity Rate |
|---|---|---|---|---|---|---|---|---|---|
| Budget-Constrained Family Customers | 445 | 45.27 | 41,008.76 | 22.45 | 7.94 | 189.29 | 0.06 | 1.43 | 0.00 |
| Premium High-Spenders | 434 | 45.84 | 75,381.76 | 44.47 | 18.84 | 1,173.50 | 0.12 | 0.36 | 0.07 |
| Young High Green-Proxy Customers | 380 | 36.57 | 27,206.77 | 49.08 | 5.91 | 105.23 | 0.28 | 0.75 | 0.08 |
| High-Recency Low-Spend Customers | 452 | 48.05 | 43,421.96 | 74.91 | 8.46 | 210.92 | 0.07 | 1.53 | 0.19 |
| High-Frequency Shoppers | 501 | 48.05 | 67,870.27 | 53.15 | 20.00 | 1,226.39 | 0.10 | 0.65 | 0.09 |

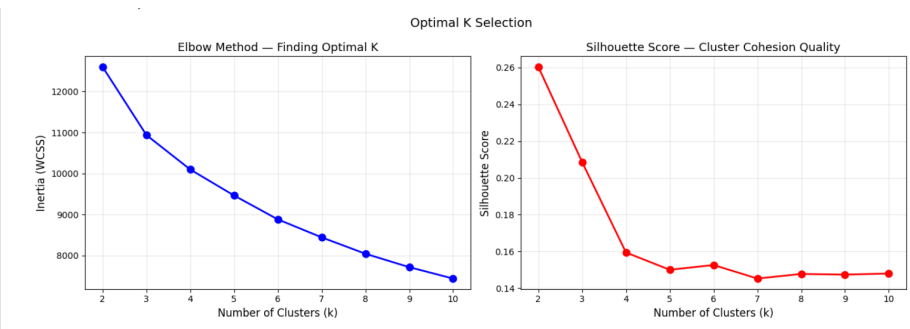



Figure 5.3: Elbow Method (Inertia) and Silhouette Score for k = 2 to 10

Figure 5.4 shows the relative size of each segment as a share of the total customer base. Figure 5.5 compares inactivity rates across segments, and Figure 5.6 compares average green-spending ratio across segments; together these indicate that the segment with the highest inactivity rate (High-Recency Low-Spend Customers) and the segment with the highest green-spending share (Young High Green-Proxy Customers) are not the same group, a point developed further in Section 5.2.1. Figure 5.7 plots Income against Monetary spend, coloured by segment, illustrating how the five segments separate along these two dimensions.

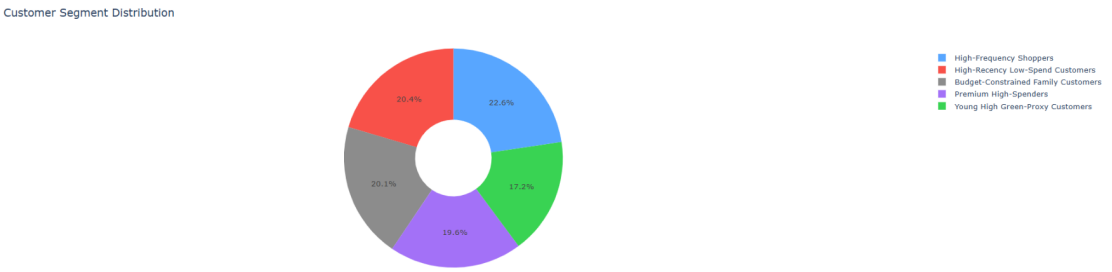



Figure 5.4: Customer Segment Distribution

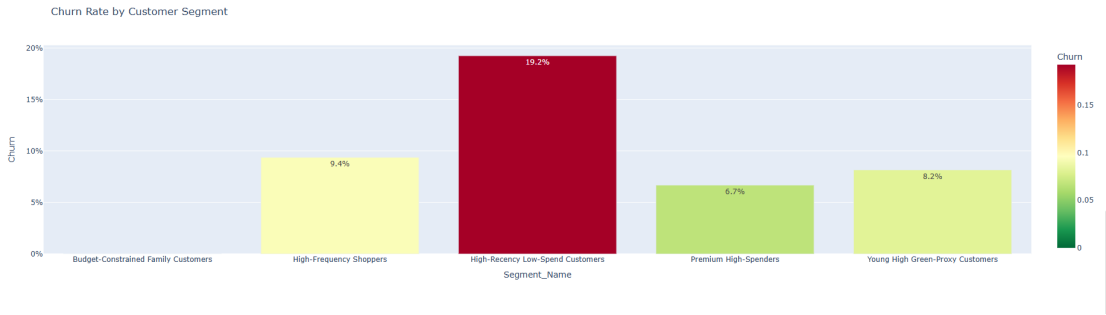



Figure 5.5: Inactivity Rate by Customer Segment

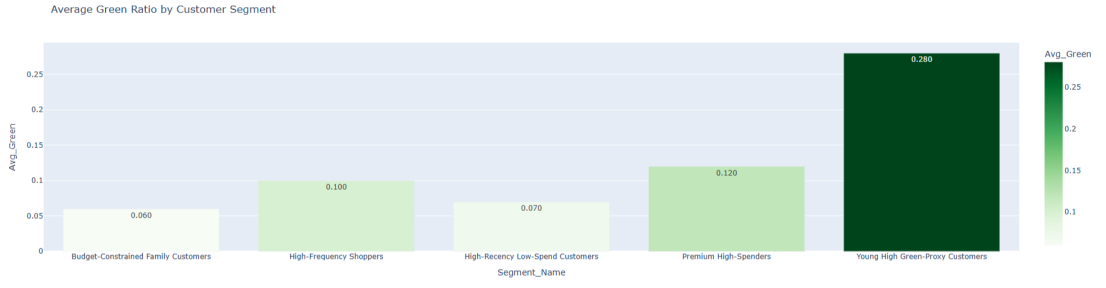



Figure 5.6: Average Green-Spending Ratio by Customer Segment

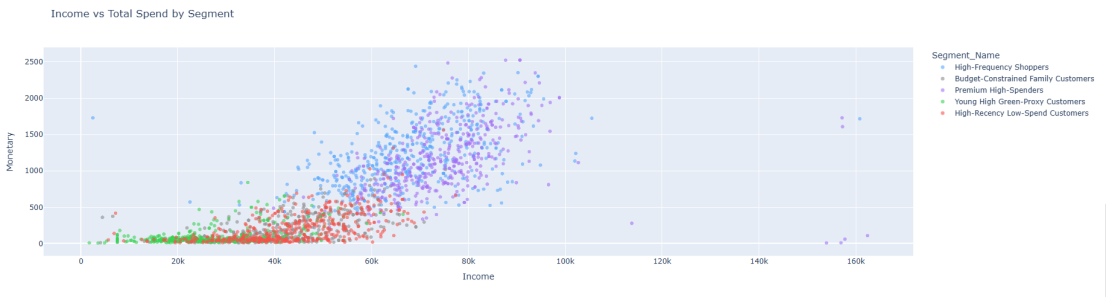



Figure 5.7: Income vs. Total Spend by Segment

### 5.1.4 Cluster Stability and Skewness Sensitivity Results

Following the stability checks described in Section 4.3.7, the five-segment solution showed moderate but imperfect stability across random initialisations (mean Adjusted Rand Index = 0.767, range 0.551–1.000 across five seeds), alongside a Davies-Bouldin Index of 2.009 and a Calinski-Harabasz score of 480.08. A comparison against a Gaussian Mixture Model produced substantially different, highly imbalanced clusters (Adjusted Rand Index = 0.115; cluster sizes ranging from 45 to 978), most plausibly reflecting GMM's distributional assumptions rather than a limitation of the K-Means solution. The skewness sensitivity check found that log-transforming Monetary produced a marginally higher silhouette score (0.162 versus 0.150) but a low level of agreement with the original clustering (Adjusted Rand Index = 0.361). The five-segment solution reported above is accordingly best understood as one reasonable, moderately stable partition of the data, rather than a definitive or uniquely correct one; this is revisited as a limitation in Chapter 6.

### 5.1.5 Comparative Model Performance

Table 5.3 compares the logistic regression baseline and the tuned XGBoost model, under both the original five-feature set and the extended eight-feature set described in Section 4.7. Under the original feature set, XGBoost achieved a test AUC of 0.5273, a marginal improvement over logistic regression's 0.5033. Hyperparameter tuning via RandomizedSearchCV (30 sampled configurations, 5-fold stratified cross-validation) selected `n_estimators`=300, `max_depth`=5, `learning_rate`=0.1, `subsample`=0.7, and `colsample_bytree`=0.9, yielding a cross-validation AUC of $0.5981 \pm 0.0482$. Under the extended feature set, XGBoost's test AUC rose to 0.5506, while logistic regression's fell to 0.4522, below the level expected by chance; this decline is consistent with logistic regression's greater sensitivity to multicollinearity, since Monetary is structurally embedded within Green_Ratio, and Frequency and Monetary are themselves correlated.

Because the test set contains only 48 positive cases, point estimates alone can be misleading; 95% bootstrap confidence intervals (1,000 resamples) were therefore computed for each AUC, along with paired bootstrap tests for the two comparisons of primary interest: whether the extended feature set improved on the original, and whether XGBoost outperformed logistic regression on the extended set. Both intervals for the individual models' AUCs, and both paired differences, are reported in Table 5.3. Neither paired comparison reached statistical significance: the extended feature set produced a higher point estimate of test AUC than the original set (mean difference = 0.0226), but the 95% confidence interval for this difference, $[-0.0567, 0.0990]$, includes zero. Similarly, XGBoost's test AUC exceeded logistic regression's on the extended feature set (mean difference = 0.0992), but this difference's confidence interval, $[-0.0301, 0.2260]$, also includes zero. Both results should accordingly be interpreted as higher point estimates rather than as statistically confirmed improvements, given the limited number of positive test cases available.
Table 5.4 reports XGBoost's classification report on the test set under the original five-feature set: precision and recall for the inactive class are 0.34 and 0.21 respectively. Under the primary evaluation criterion (test-set AUC), XGBoost produced the higher point estimate under both feature sets, though all models performed only marginally above the 0.500 random-classifier baseline, and the improvements from adding features or switching models were not statistically distinguishable from chance variation given the available sample size.

Table 5.3: Model Performance Summary, Including Bootstrap 95% Confidence Intervals

| **Metric** | **Value** | **95% CI** |
|---|---|---|
| Logistic Regression – Test AUC (original 5-feature set) | 0.5033 | – |
| XGBoost – Cross-Validation AUC (tuning, original set) | 0.5981 ± 0.0482 | – |
| XGBoost – Test AUC (original 5-feature set) | 0.5273 | [0.4227, 0.6361] |
| Logistic Regression – Test AUC (extended 8-feature set) | 0.4522 | – |
| XGBoost – Test AUC (extended 8-feature set) | **0.5506** | [0.4588, 0.6488] |
| AUC Difference: Extended − Original (XGBoost) | +0.0226 | [-0.0567, 0.0990] |
| AUC Difference: XGBoost − Logistic Regression (extended) | +0.0992 | [-0.0301, 0.2260] |

Table 5.4: XGBoost Classification Report on Test Set, Original Five-Feature Set (n = 553)

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| Not Inactive (0) | 0.93 | 0.96 | 0.94 |
| Inactive (1) | 0.34 | 0.21 | 0.26 |

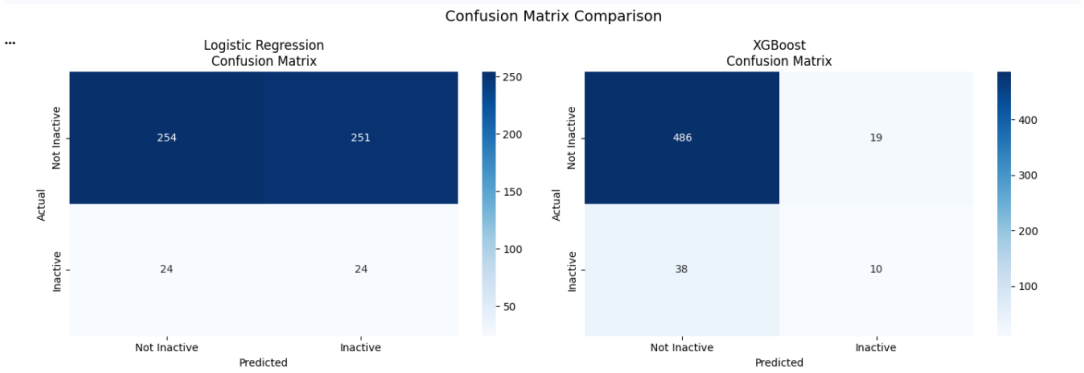



Figure 5.8: Confusion Matrices: Logistic Regression vs. XGBoost (Original Five-Feature Set, Test Set)

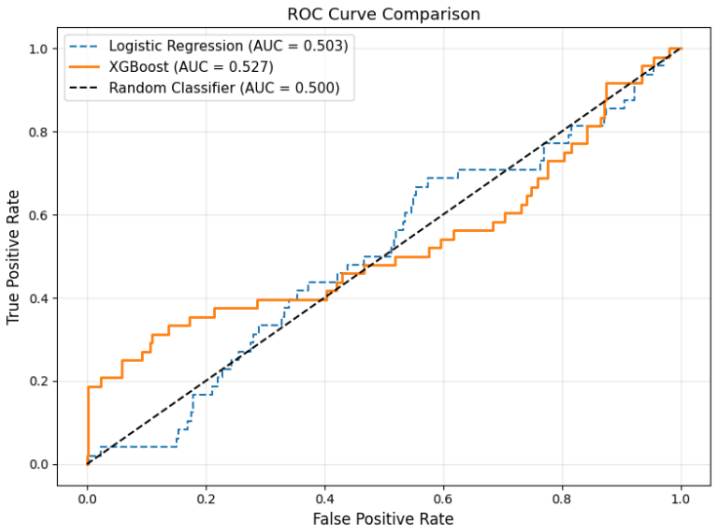



Figure 5.9: ROC Curve Comparison (Original Five-Feature Set)

### 5.1.6 SHAP Explainability Results

SHAP was computed on the extended eight-feature XGBoost model described in Section 5.1.5, as this was the version carried forward for interpretation. The global SHAP ranking places Income first, followed by Age, Green_Ratio, Monetary, Days_As_Customer, Frequency, Total_Kids, and Total_Accepted_Campaigns (Figures 5.10, 5.11), with

mean absolute SHAP values of 0.967, 0.757, 0.730, 0.667, 0.662, 0.335, 0.180, and 0.061 respectively. Notably, Green_Ratio's importance changed only modestly once Monetary and Frequency were reintroduced into the model, despite both representing plausible sources of confounded signal; this pattern, together with the multivariate logistic regression results below, is examined further in Section 5.2.3.

Table 5.5 reports mean absolute SHAP values by feature and segment. Age has the largest single value in the table, within the Premium High-Spenders segment (1.002). Within the Young High Green-Proxy Customers segment, Green_Ratio reaches its highest importance across all five segments (0.865), though this remains below the largest value observed for Age or Income in other segments.The dependence plots (Figures 5.12, 5.13) show substantial colour variation by Income at similar Green_Ratio and tenure values. The individual waterfall plot (Figure 5.14) shows one example customer, from the High-Frequency Shoppers segment, with a predicted inactivity probability of 0.0085, correctly classified as not inactive.

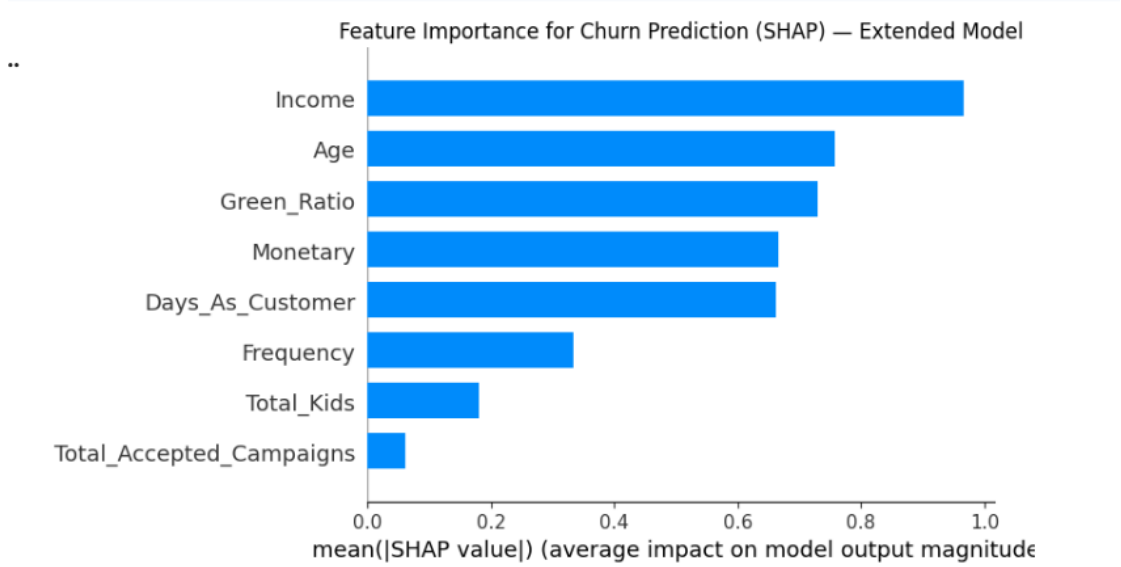



Figure 5.10: Global Feature Importance (Mean Absolute SHAP Value), Extended Model

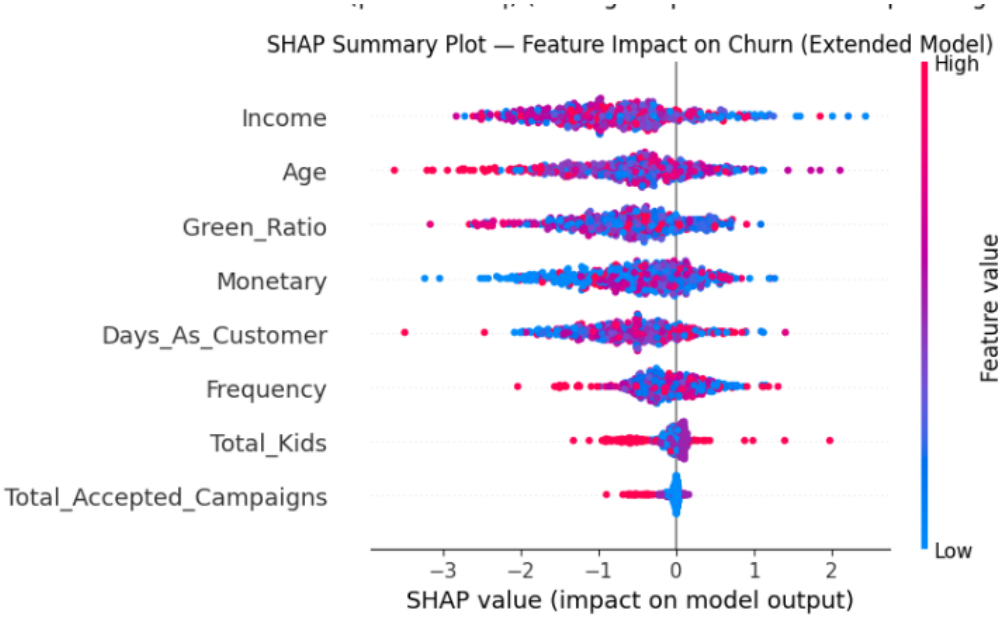



Figure 5.11: SHAP Beeswarm Plot, Extended Model

Table 5.5: Mean Absolute SHAP Value by Feature and Segment, Extended Model

| Segment | Income | Age | Green_Ratio | Total_Kids | Days_As_Customer | Frequency | Monetary | Campaigns |
|---|---|---|---|---|---|---|---|---|
| Budget-Constrained Family Customers | 0.880 | 0.604 | 0.683 | 0.303 | 0.615 | 0.325 | 0.729 | 0.025 |
| High-Frequency Shoppers | 1.099 | 0.724 | 0.691 | 0.097 | 0.516 | 0.372 | 0.532 | 0.093 |
| High-Recency Low-Spend Customers | 0.958 | 0.761 | 0.642 | 0.298 | 0.724 | 0.311 | 0.674 | 0.032 |
| Premium High-Spenders | 0.971 | **1.002** | 0.813 | 0.104 | 0.739 | 0.377 | 0.495 | 0.120 |
| Young High Green-Proxy Customers | 0.889 | 0.726 | 0.865 | 0.087 | 0.771 | 0.278 | 0.953 | 0.032 |

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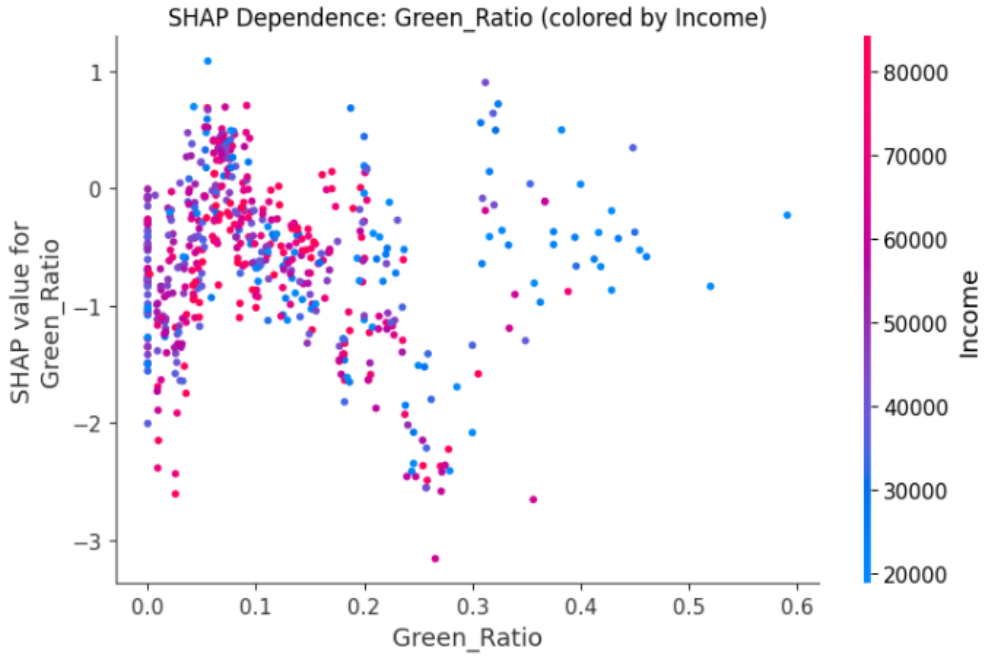



Figure 5.12: SHAP Dependence Plot: Green_Ratio (Coloured by Income), Extended Model

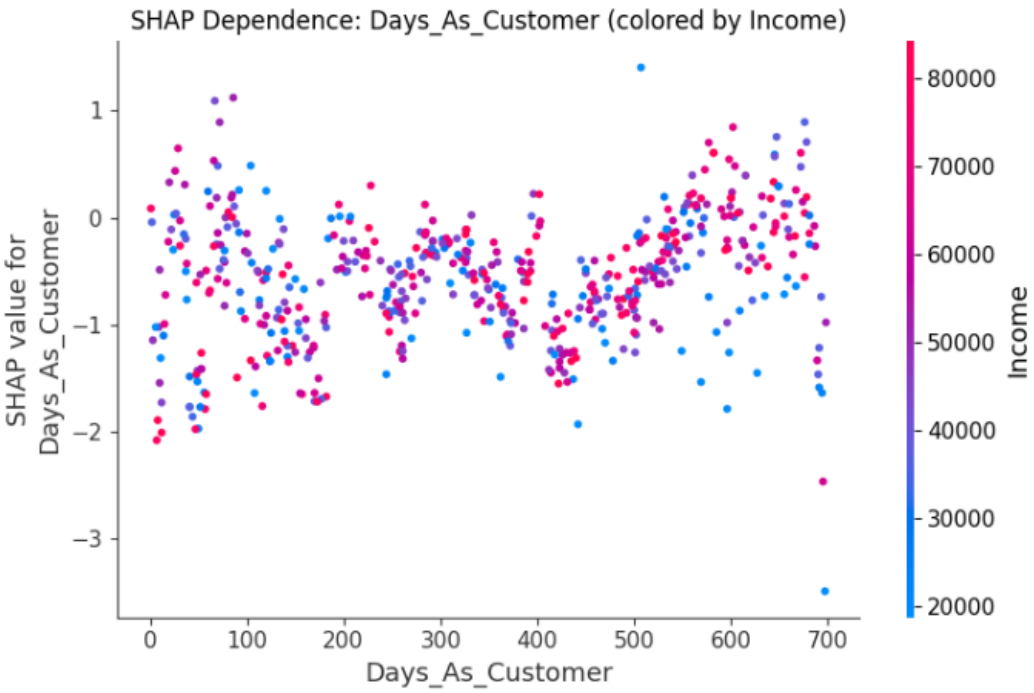



Figure 5.13: SHAP Dependence Plot: Days_As_Customer (Coloured by Income), Extended Model

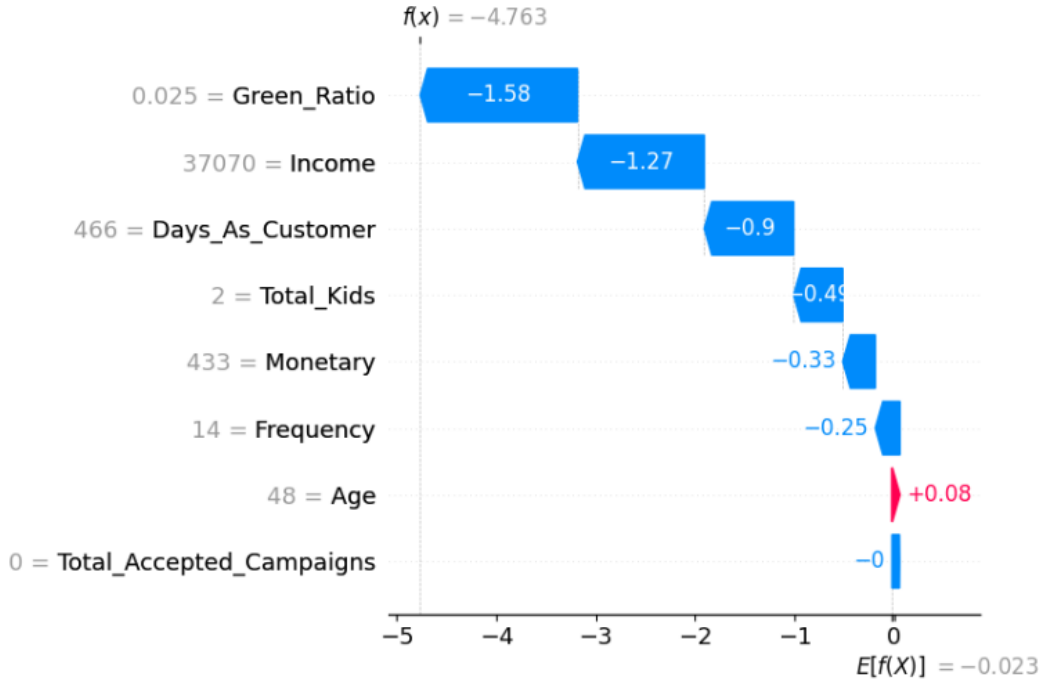



Figure 5.14: Individual SHAP Waterfall Plot for an Example Customer

## 5.2 Discussion

### 5.2.1 Interpreting the Segmentation Results

The five segments indicate that green-spending share and inactivity are concentrated in different customer groups rather than the same one: Young High Green-Proxy Customers have the highest green-spending ratio (0.28) but a below-average inactivity rate (8%), while High-Recency Low-Spend Customers have the highest inactivity rate (19%) but a below-average green ratio (0.07). This pattern is broadly consistent with the Business Question 2 finding and the formal statistical tests in Section 5.1.2, indicating that green spending and inactivity are not meaningfully linked in this dataset. It should not, however, be interpreted as a second, independent confirmation of that result: Recency is one of the eight features used to form these clusters, and Recency also defines the inactivity label by construction, so any segment with elevated Recency will tend mechanically toward elevated inactivity regardless of its other characteristics. The chi-square test, logistic regression, and nonlinearity check reported in Section 5.1.2 remain the primary evidence for the absence of a green-spending/inactivity relationship; the segmentation is reported here descriptively, as a means of characterising the customer base, rather than as additional statistical proof.

The choice of $k = 5$ over the statistically optimal $k = 2$ also warrants justification. A silhouette-maximising $k = 2$ would separate customers only into a coarse high-value/low-value split, insufficiently granular to support differentiated marketing action; $k = 5$ sacrifices some cluster cohesion in exchange for personas of greater practical use to a marketing team. This reflects the general point made in Chapter 2 that the statistically optimal clustering solution is not always the most useful one for a business audience. The cluster stability and skewness checks reported in Section 5.1.4 qualify this further: given moderate but imperfect stability across random seeds (mean ARI = 0.767) and demonstrated sensitivity to how Monetary is transformed (ARI = 0.361 against a log-transformed alternative), the five-segment solution should be understood as one reasonable, defensible partition of the customer base rather than a definitive one.

### 5.2.2 Interpreting Model Performance

All models perform only marginally above chance under both feature sets (original XGBoost test AUC = 0.5273; logistic regression = 0.5033; extended XGBoost = 0.5506; extended logistic regression = 0.4522). This is best explained by the leakage

correction described in Chapter 4: because the inactivity label is defined purely from Recency, and Recency must be excluded as a feature, comparatively little linear signal remains among the remaining candidate features, a pattern confirmed by the correlation results in Figure 5.1, where no feature exceeds $|r| = 0.03$ with the label. The weak performance observed is therefore a largely expected consequence of removing leakage, rather than a modelling deficiency.

The gap between the cross-validation AUC ($0.5981 \pm 0.0482$) and the original test AUC (0.5273) is similarly attributable to sample size: with only 194 inactive customers overall, and approximately 146 of these within the training set, each cross-validation fold contains approximately 29 positive examples, rendering the cross-validation estimate unstable and optimistic relative to the held-out test result. Severe class imbalance of this kind is a commonly cited source of unreliable performance estimation within churn-prediction research more broadly (Jain et al. 2020). The bootstrap analysis reported in Section 5.1.5 reinforces this point directly: neither the improvement associated with the extended feature set, nor XGBoost's advantage over logistic regression, was statistically distinguishable from zero, given the limited number of positive test cases. The test-set AUC, and specifically its confidence interval rather than the point estimate alone, is accordingly the figure that should be regarded as authoritative, and both point estimates should be interpreted as suggestive rather than conclusive.

### 5.2.3 Reconciling the SHAP Findings with the EDA Findings

The SHAP results (Section 5.1.6) show the extended model assigning measurable importance to Green_Ratio, particularly within the Young High Green-Proxy Customers segment (0.865), which appears, at first inspection, to sit in tension with the near-zero correlation and non-significant statistical tests reported in Sections 5.1.1 and 5.1.2. Because the XGBoost model's overall AUC (0.5506 under the extended feature set) is only marginally better than chance, its internal feature-importance rankings should be interpreted as exploratory rather than as independent evidence of a genuine effect; a model of limited discriminative power can assign a feature substantial internal weight without that weight reflecting a genuine pattern among the underlying customers.

A plausible hypothesis, and the one initially considered in this thesis, is that Green_Ratio's SHAP importance reflects confounding: Green_Ratio may be statistically entangled with Income or Age, either of which could independently relate to inactivity, causing the model to rely on Green_Ratio as an indirect proxy. This hypothesis was

tested directly using the multivariate logistic regression described in Section 4.5.1, predicting inactivity from Green_Ratio together with Income, Age, Total_Kids, and Days_As_Customer. Green_Ratio remained non-significant after these controls were introduced ($\beta = -0.034$, odds ratio = 0.967, 95% CI [0.820, 1.139], $p = 0.684$), and, notably, none of the control variables reached significance either (all $p > 0.05$); the overall model did not fit significantly better than an intercept-only model (likelihood-ratio $p = 0.705$). This result does not support a straightforward confounding explanation, since confounding requires the confounding variable itself to carry a genuine relationship with the outcome, which was not observed here for Income, Age, or any other control examined. The more parsimonious interpretation is that the model's reliance on Green_Ratio reflects the general absence of predictive signal across the available feature set once Recency is excluded (Section 5.2.2), rather than a specific, identifiable confound. The dependence plots (Figures 5.12, 5.13), which show substantial colour variation by Income even among customers with similar Green_Ratio or tenure values, are consistent with this weak-signal interpretation rather than with a clear confounding pattern, since a genuine confounder would be expected to demonstrate a more systematic relationship with the outcome than was observed here.

This finding is broadly consistent with the wider consumer-behaviour literature on the "green gap," which documents a persistent disconnect between sustainability attitudes and actual purchasing outcomes (ElHaffar et al. 2020). This thesis extends that observation further: not only do stated environmental attitudes frequently fail to predict actual purchasing behaviour, but in this dataset, observed green purchasing behaviour itself failed to predict a further downstream outcome, customer inactivity, even under a series of tests designed to detect a subtler, indirect relationship.

**Conclusion for stakeholders:** across a raw correlation, a chi-square test, linear and nonlinear logistic regressions, and a multivariate regression controlling for plausible confounders, no evidence was found that green-spending behaviour is related to customer inactivity. Green_Ratio should not be used as a retention-risk signal, and the importance a model assigns to it via SHAP should not be interpreted as confirmation that it is substantively meaningful.

### 5.2.4 Practical Implications

The findings carry tentative implications for marketing strategy, presented here as directions warranting further testing rather than as validated recommendations, since

none of the following has been evaluated experimentally. Because green spending and inactivity showed no measurable relationship in this dataset, green marketing and retention marketing could reasonably be planned as two separate efforts rather than a single combined lever. Segment-specific possibilities follow descriptively from the discussion above:

- Premium High-Spenders show high spending power but only moderate green-spending share (0.12); this could be regarded as a candidate opportunity for green upselling, distinct from any retention concern, since this segment's inactivity rate (7%) is already close to the sample average.
- Young High Green-Proxy Customers show a green-spending share well above the other segments (0.28) despite lower income; affordable green bundles or loyalty incentives may represent a more suitable starting point than premium pricing, though this has not been tested.
- High-Recency Low-Spend Customers show the highest inactivity rate observed (19%), making this the segment where retention attention would plausibly matter most. Given the model's limited discriminative power, win-back activity would be better triggered directly by the Recency threshold itself rather than by the model's predicted probability, which should be treated as, at best, a secondary, low-confidence signal.
- Budget-Constrained Family Customers show a 0% inactivity rate in this sample. This reflects that no customers in this segment crossed the study's 90-day threshold within the observed period, rather than confirmed evidence that this segment carries no genuine retention risk; marketing spend directed at this group might reasonably prioritise growing green share-of-wallet over retention activity, though this too remains a hypothesis rather than a demonstrated conclusion.

These points are best understood as testable hypotheses for future segment-targeted experimentation (Chapter 6), rather than as strategies this thesis has demonstrated to be effective.

### 5.2.5 Answering the Research Questions

- **RQ1** (Segmentation): Answered. Five behaviourally distinct segments were produced (Section 5.1.3), though stability testing showed only moderate reproducibility across random initialisations (mean ARI = 0.767) and some sensitivity to feature skewness (Section 5.1.4); the segmentation is accordingly

best treated as one reasonable partition of the data rather than a uniquely determined one.

- **RQ2** (Green spending and inactivity): Answered. No statistically significant relationship was found across a chi-square test, a linear logistic regression, and a nonlinearity check (Section 5.1.2).

- **RQ3** (Predictability of inactivity): Answered. Inactivity is only marginally predictable once leakage is removed, under both the original and extended feature sets, and bootstrap testing found neither the extended feature set nor XGBoost's advantage over logistic regression to constitute a statistically significant improvement (Section 5.1.5, 5.2.2).

- **RQ4** (SHAP-based drivers): Answered. Income, Age, and Green_Ratio carry the largest SHAP importance in the extended model, but a multivariate logistic regression found no significant relationship between Green_Ratio and inactivity even after controlling for Income, Age, Total_Kids, and Days_As_Customer, and none of the control variables were significant either; this does not support a confounding explanation, and instead indicates that the model is detecting weak, non-generalisable patterns consistent with its near-chance overall performance (Section 5.1.6, 5.2.3).

- **RQ5** (Segment-specific strategy): Answered. Several segment-specific directions follow descriptively from the segmentation and inactivity results, framed as testable hypotheses for future experimentation rather than validated recommendations, given that none have been evaluated directly (Section 5.2.4).

# Chapter 6 Conclusions

## 6.1 Conclusions

This thesis examined a question frequently raised in marketing practice but rarely tested directly: are customers who purchase more sustainably also less likely to lapse? Using a dataset of just over two thousand retail customers, this study developed a complete pipeline, comprising data cleaning, feature engineering, segmentation, a model of purchase inactivity, and explainability, specifically to examine this assumption empirically rather than accept it as given.

Several findings merit particular attention. The first concerns the construction of the inactivity label itself, which proved more consequential than initially anticipated. The original label combined purchase recency with campaign response, and because campaign response was also intended as a model feature, this construction risked allowing the model to reconstruct its own label rather than learn genuine behavioural signal (Kaufman et al. 2011). Identifying this issue and rebuilding the label from recency alone was a necessary step for the credibility of the analysis that followed, and subsequent feature-engineering decisions, including the corrections to the Frequency and Green_Ratio calculations, were held to the same standard.

Second, the segmentation produced five behaviourally distinct customer groups, ranging from a budget-constrained family segment with no observed inactivity in this sample, to a smaller, younger, lower-income group whose green-spending share exceeded that of every other segment. These groups are usable and business-interpretable, although stability testing indicated that the five-segment structure is moderately, rather than perfectly, reproducible, and exhibits measurable sensitivity to how skewed variables were transformed. This limitation is reported directly rather than omitted, and is revisited under Limitations below.

Third, and central to this thesis's empirical contribution, no evidence was found to support the assumed relationship between green spending and loyalty, across any

of the tests applied. Inactivity rates varied minimally across low, medium, and high green-spending customers, the correlation between the two was close to zero, and a chi-square test, a linear logistic regression, a nonlinearity check, and a multivariate regression controlling for income, age, household size, and tenure each failed to identify a statistically significant relationship. This constitutes a well-supported answer to the thesis's central research question, notwithstanding that it contradicts an assumption common in marketing practice; a negative result substantiated through this range of testing does not represent a weaker finding than a positive one, only a less convenient one.

Fourth, once the leakage was corrected, predicting inactivity proved genuinely difficult under both the original and extended feature sets, and bootstrap testing indicated that neither reintroducing additional features nor substituting XGBoost for logistic regression produced a statistically confirmed improvement in performance. This does not reflect a deficiency in the modelling approach; rather, it reflects the limited signal that remains once the variable defining the label by construction is properly excluded, and is reported here as a substantive finding rather than a limitation to be minimised.

Fifth, SHAP was used to examine an apparent inconsistency: the model assigned measurable importance to Green_Ratio despite the absence of any relationship in the raw data. Testing this directly through a multivariate regression found that neither Green_Ratio nor any candidate confounding variable reached statistical significance, a result inconsistent with a specific confounding explanation and more consistent with the model detecting weak, non-generalisable patterns in keeping with its limited overall discriminative performance. Reaching this more precise conclusion required examining the model's internal behaviour through SHAP and subsequently testing the resulting hypothesis formally, rather than accepting the model's output uncritically or dismissing it without further examination.

Taken together, this study provides a direct, evidence-based response to a widely held marketing assumption: within this dataset, using the green-spending proxy and inactivity definitions described in Chapter 4, no meaningful association was identified between sustainability-oriented purchasing and customer inactivity. Within the scope of this analysis, sustainability and retention appear to represent separate concerns rather than a single, combined one.

## 6.2 Limitations

Several limitations warrant explicit acknowledgment. The dataset, while suitable for this study, represents a single cross-sectional snapshot rather than a longitudinal record of customer behaviour, which constrains the extent to which any finding here can be characterised as causal rather than associative. The dataset is also of moderate size; with only 194 customers meeting the inactivity threshold, both the clustering and the predictive model were working with a limited number of positive examples, a constraint reflected directly in the instability of the cross-validation results and the width of the bootstrap confidence intervals reported in Chapter 5. The clustering itself, while producing interpretable segments, demonstrated only moderate stability across random initialisations and measurable sensitivity to the transformation applied to skewed features such as Monetary; the five-segment structure should therefore be regarded as one reasonable partition of the data rather than a definitive one. The operationalisation of "green" behaviour employed here, based on two of six available product categories, represents a comparatively narrow interpretation of sustainability; a more granular, product-level environmental scoring system might yield different conclusions (Clark & Tilman 2017). Finally, the ninety-day inactivity threshold, while a reasonable and defensible choice, was not the only threshold that could have been adopted, and an alternative choice might alter some of the reported figures.

## 6.3 Future Works

Several directions for future research follow from the constraints of this study. Extending the analysis from a single snapshot to a longitudinal dataset would represent a particularly valuable next step, enabling the relationship between green spending and inactivity to be tested for a lagged rather than contemporaneous effect, and allowing inactivity to be modelled as a proper survival process rather than a static label. Incorporating richer behavioural data, such as website activity, customer service interactions, or return histories, would provide a predictive model with more substantial input than was available in this study, and may improve predictive performance beyond the near-chance level observed here. With respect to the sustainability measure, developing a more granular, product-level environmental scoring system, rather than the two-category proxy adopted in this thesis, would provide a means of testing whether the null result reported here is specific to this narrow definition of "green" or holds under a broader operationalisation (ElHaffar et al. 2020). Finally, the segment-specific directions proposed in Chapter 5 remain

untested hypotheses rather than validated strategy; evaluating them directly, for example through an A/B test offering green product bundles to the Premium High-Spenders segment, would establish whether these directions hold in practice rather than only in principle.

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